Development of a Smart Barangay Kiosk System with Offline-First Document Request Handling and Resident Management

by

Dennis Leonardo S. Cuadro, BS Computer Engineering
Karl S. Ignacio , BS Computer Engineering
Frank John Paul L. Tresvalles, BS Computer Engineering

A Thesis Report Submitted in Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Science in Computer Engineering

Ateneo de Naga University

2026

APPROVAL SHEET

This is to certify that we have supervised the preparation of and read the senior project paper entitled Development of a **Smart Barangay Kiosk System with Offline-First Document Request Handling and Resident Management** and that the said paper has been submitted for final examination by the Oral Examination Committee

Engr. Karla Louissa Marie D. Sobrevilla
Academe Adviser

As members of the Oral Examination Committee, we certify that we have examined this paper and hereby recommend that it be accepted as fulfillment of the senior's project required for the degree of Bachelor of Science in Computer Engineering.

Engr. Elmer C. Claveria Jr.
Panel Member 1

Engr. Joel R. Noche
Panel Member 2

Engr. Carlo Giovanni E. Olivan
Panel Member 3

This senior project paper is hereby approved and accepted by the College of Engineering as the fulfillment of the senior's project requirement for the degree of Bachelor of Science in Computer Engineering.

Engr. Celeste M. Ojeda
Department Chair

Engr. Referendo D. Soriano
College Dean

ACKNOWLEDGMENT

The researchers would like to give thanks and appreciation to all individuals who offered their assistance and support, without which this work would not have been possible.

The researchers wish to express their deepest appreciation to their adviser Engr. Karla Louissa Marie Sobrevilla who provided them with continuous support and valuable guidance. Her mentorship and guidance have been truly instrumental in shaping the direction and success of this study.

The researchers express their sincere appreciation to panelists Engr. Elmer Claveria Jr. and Engr. Joel Noche and Engr. Carlo Giovanni Olivan who dedicated their time to share their knowledge and provide critical evaluation of the research.

In addition, the researchers extend their sincere thanks to the Barangay Captains of Barangay Sta. Maria and Barangay Maangas, together with their staff, for their cooperation, accommodation, and assistance during the data gathering phase. The researchers required their assistance because they needed reliable data to successfully complete the study.

Finally, The researchers want to express their most sincere thanks to their family members who helped them throughout their research work. The research team achieved their study goals through their dedicated efforts which included both hard work and enduring dedication together with their continuous drive that helped them achieve exceptional results.

ABSTRACT

The Smart Barangay Kiosk System is an offline-first Progressive Web Application (PWA) developed to modernize document request handling and resident management in Philippine barangays. The system uses React 19 together with Vite and Supabase and TailwindCSS to create a self-service kiosk interface that runs on tablets which feature Bluetooth thermal printers and administrative dashboards for barangay officials and resident portals that allow remote document requests with complete offline functionality and a super-administrator panel for managing multiple barangays. The system supports eleven barangay document types—Barangay Clearance, Certificate of Indigency, Certificate of Residency, Certificate of Good Moral Character, Certification for First Time Job Seeker, Certification for Monthly Income, Certification for Solo Parent, Certification PDA, Business Clearance, Barangay ID, and Barangay Certification—with a custom "Other documents" option, automatic control-number generation, receipt printing and SMS notifications through PhilSMS while providing users full access to both kiosk and resident portal functions without the need for internet through IndexedDB caching and Workbox service worker technology. Both the kiosk terminal and the resident portal allow document request submission without an internet connection; requests are queued locally and synchronized automatically when connectivity is restored. The system went through its initial testing phase in Barangay Maangas and Barangay Sta. Maria, which are located in Presentacion, Camarines Sur. Assessment results showed that the system delivered three major benefits by decreasing document processing time while making services more accessible to residents and improving administrative operations, which proved that rural local governance can successfully implement low-cost offline digital solutions.

Keywords: smart barangay kiosk, offline-first PWA, document request, resident management, e-governance, kiosk

TABLE OF CONTENTS

APPROVAL SHEET i
ACKNOWLEDGEMENT ii
ABSTRACT iii
TABLE OF CONTENTS iv
LIST OF TABLES v
LIST OF FIGURES vi
CHAPTER 1: Introduction **1**
1.1. Background of the Study 1
1.2. Statement of the Problem 5
1.3. Significance of the Study 6
1.4. Objectives 7
1.5. Scope and Delimitations 8
CHAPTER 2: Review of Related Literature and Studies **9**
2.1. Theoretical Basis
2.1.1. E-Governance
2.1.2. Offline-First Progressive Web Apps (PWAs)
2.1.3. HCI for Low-Literacy Users
2.1.4. Database Management and BaaS
2.1.5. Tablet-Based Kiosk Deployment
2.1.6. System Security and SMS Integration
2.2. Related Studies
2.2.1. Community Data Management Using IoT Devices
2.2.2. Design and Implementation of a Local E-Government Portal
2.2.3. Barangay Management Information System
2.2.4. Web-Based Information Management System
2.2.5. Development of an Offline-Capable Health Management System
2.2.6. E-Barangay Portal with SMS Integration
2.2.7. Enhanced Barangay Information Residency Certificate Issuance System
2.2.8. Mobile and Web Barangay System
2.2.9. Barangay Certificate Issuance and Record Management System with Descriptive Analytics
2.2.10. A Risk Assessment of a Barangay Management System Portal
2.3. Synthesis and Research Gap
CHAPTER 3: Methodology
3.1. Research Design and Systems Model
3.2. Requirement Analysis and Data Collection Methods
3.2.1. Interviews and Surveys
3.2.2. Observation
3.2.3. Document Analysis
3.3. System Design, Development and Implementation Tools and Tecnologies
3.3.1. Hardware Components
3.3.2. Software and Development Tools
3.3.3. System Architecture and Module Design
3.3.4. System Flowchart

3.3.5. Database Design
3.3.6. Offline-First PWA Behavior
3.4. Project Planning and Execution
3.4.1. System Features Based on Objectives
3.4.2. Project Cost Estimation and Hardware Breakdown
3.4.3. Project Timeline and Development Schedule
3.5. Testing, Evaluation, and Data Analysis
3.5.1. Resident Pre-Usage Survey Results
3.5.2. Resident Post-Usage Survey Results
3.5.3. Admin Pre-Usage Survey Results
3.5.4. Admin Post-Usage Survey Results
3.5.5. User Feedback Analysis
3.5.6. Summary of Evaluation Results
3.6. Target Respondents and Sampling
3.6.1. Barangay Officials and Staff
3.6.2. Barangay Residents
3.6.3. Sampling Technique
3.6.4. Respondent Profile and Sample Size
3.7. Development Boundaries and Prototype Constraints
3.8. Ethical Considerations
CHAPTER 4: Conclusion
CHAPTER 5: Recommendations
REFERENCES
APPENDIX A
APPENDIX B

CHAPTER 1

INTRODUCTION

1.1. Background of the Study

Local Government Units (LGUs) in the Philippines have been on the front line in promoting digital transformation in public service delivery in recent years [1]. On a more intimate level, the barangay is the smallest unit implementing the expected government functions. Most barangays still process documents, organize files, and manage resident information in manual and paper-based systems. These conventional methods, though familiar, are often inefficient and time-consuming, especially during peak seasons when the demand for barangay documents significantly increases

Digitalization of service delivery, however, has its disadvantages. Services are usually slow, apathetic, and painfully long, especially during the peak seasons. Residents sometimes have to wait for hours to process a simple document request.[2]. The current system is inconvenient, causing frustration for senior citizens, persons with disabilities, and those short on time due to work or other personal obligations.

A number of residents are unable to access some essential barangay services because of poor digital infrastructure and internet connection within their respective localities [3]. Urban areas might benefit from high-tech internet services, while poor rural barangays are thus unable to participate in digital programs [4]. While some municipalities already share online portals, this cannot be assumed to guarantee that most residents, especially those in rural areas, own smartphones, computers, or stable internet connections [5]. Hence, these services will actually exclude them from the government's e-Governance scheme. Instead of being inclusive, digital

initiatives may further widen the gap between digitally connected and disconnected communities [6].

The effective use of low-cost and accessible technologies has enabled the development of affordable, simple, and efficient kiosks for community service delivery [7]. These systems provide promising opportunities to create digital platforms that work even in areas with limited power and network connectivity. The Smart Barangay Kiosk System allows residents to request documents and barangay officials to manage resident records without requiring residents to own a personal device or maintain an internet connection. The kiosk acts as a digital gateway for the community- easy to use, always available, and not dependent on the residents' access to technology.

In addition to being accessible, this will also simplify administrative processes, lessen the load on staff, and encourage transparency in the community . Not only will staff spend less time manually managing the walk-in requests but they could also carry out actions such as verifying, approving, and updating records by using the system. It also will keep all transactions digitally recorded and reduces the chances of misplacing paper documents or having inconsistent data.

Thus, the present study focuses on the development of an integrated Smart Barangay Kiosk System for document request handling and resident management; a step forward in the practical direction of modernizing barangay operations in consonance with national goals of digital transformation. The use of a tablet-based kiosk with a Bluetooth thermal printer in this context emphasizes low-cost innovation that suits the needs and capabilities of local communities [7].

Ever since, municipalities have been enhancing web platforms through which residents may request documents. On the other hand, for those living on the wrong side of the digital divide especially in rural setups or barangays of low-income portals and towns is a big challenge, thereby posing an obstacle for citizens to access essential public services [3]. Digital transformation must consider the availability of both infrastructure and user access. In many barangays, unfortunately, the residents are limited by not only weak connectivity but also the issue of personal digital devices and computer literacy [5]. In the end, many residents are unaware or unable to use online platforms, despite its availability in their area.

It becomes clear that there will be an immediate need for a localized, offline, and very simple-to-use platform through which every resident can access barangay services, taking into account their level of technology access. The need is not just for innovation, but for inclusive innovation—one that reaches every member of the community, regardless of their digital limitations [6].

The Smart Barangay Kiosk System operates under the current conditions by digitalizing essential barangay processes through its touchscreen tablet kiosk which operates from the barangay hall. The system enables residents to request documents through its self-service system which requires no personal device usage. The kiosk is installed in a public and easily accessible location, allowing even the least tech-savvy individuals to navigate basic services.

The kiosk will also have a remote management administrative dashboard that enables authorized barangay officials in the management of resident records and processing of document requests. This dashboard will allow officials to have real-time information about managing more effectively citizen requests and also manage timely and accurate records. Improvement in quality

is not limited to the service area alone, but has also been made for accountability and transparency in public administration [8].

The system will operate in an offline-first manner and integrates into rural barangays with the added advantage of operating effectively even under conditions with very limited or no internet connectivity for long periods. When internet access becomes available, the system will automatically synchronize data with the server, keeping everything updated without needing to upload data manually. This two-way mode (offline and online) increases the reliability of the system.

The combination of all these features with the use of a consumer-grade Android tablet and a portable Bluetooth thermal printer makes this solution cost-effective, scalable, and sustainable for community use. This system can be adopted and maintained by small barangays without large budgets or extensive technical staff. Its compact and energy-efficient properties make it an ideal system for remote areas with poor or unreliable power supply.

The Smart Barangay Kiosk System aims to enhance access to document services and secure resident data management that will go well with offline functionality to promote that kind of digital inclusion within the community. Through this, technology is also given a role to play in local community efforts toward transparency, access, and delivery of services [8]. By transforming how barangays interact with their constituents, this innovation contributes not only to improved service delivery but also to building trust and participation among residents, strengthening the relationship between the government and the people it serves.

1.2. Statement of the Problem

Manual and repeated processes will continue to undermine the efficiency of barangay offices. Therefore, the residents must go to the barangay hall to request documents. Any processing time is slow, so documents are often misfiled. Certainly, this age-old system will continue to restrict access to handling documentation, especially for the elderly or those unable to travel for work during regular business hours.

Moreover, the system currently lacks a central digital platform which would enable storage and management of resident data. Document tracking becomes challenging because the system lacks document request information and document verification capabilities. The system requires residents to visit official offices for document requests because officials have established minimal contact with them.

This will continue causing slow document processing, inefficient management of the residents' data, and being less transparent, as well as making already difficult access to exclusive digital platforms worse among residents who do not have personal devices or reliable internet access. The absence of a central system for request handling, transaction tracking, or timely availability of public information causes an access inconsistency in service delivery while the administrative workload remains heavy.

That is why there is a call for a cost-efficient, less-power dependent, and scalable solution that can digitize core services of barangays and run even in less connectivity. Such a solution should also improve both resident satisfaction and administrative efficiency when it comes to resolving administrative requests. Additionally, challenges related to system deployment, adoption, and maintenance must be addressed to ensure sustainability at the barangay level.

1.3. Significance of the Study

The Smart Barangay Kiosk System provides a serious benefit to the lives of residents and barangay councilors alike. By utilizing cheap and available digital technologies, the system will simplify some of the more common administrative tasks causing inefficient transactions and perhaps at least attempt to narrow the digital divide within the community[6].

This innovation will not only simplify service delivery but also promotes transparency, accountability, and inclusivity in local governance. The benefits can be categorized as follows:

For Residents:

- Setting up a user-friendly, freely accessible kiosk that would allow residents to request papers any time they wish without needing to use a computer or mobile phone would be very helpful.
- It will greatly decrease the time and effort required by residents to obtain barangay services.
- Will improve inclusiveness and digital literacy, especially among low-income and elderly communities.

For Barangay Officials:

- Reduced manual labor as a result of centralized record administration and automated processing.
- Issuance of records that are more accountable, transparent, and accurate.
- This enables remote data management and monitoring via a secure admin dashboard.

For Local Government Unit (LGU):

- This supports LGUs' digital transformation initiatives through a scalable, replicable, and cost-effective service delivery model.
- It enables centralized transaction logging and increased accessibility of resident information in enhancing governance efficiency.
- It enhances compliance with national policies on e-governance and modernization programs.

For Future Researchers:

- The developmental foundation work now leads to better community service kiosks together with IoT-based governance tools.
- The system expansion potential includes biometric authentication, mobile system integration, AI analytics development and multiple barangay system implementation.
- The research funding supports the development of offline-first systems while creating smart communities and localized e-governance solutions.

1.4. Objectives

The main objective of the study is to develop, design, and implement a Smart Barangay Kiosk System that will modernize and digitize the way barangay services are delivered to the community. With the use of cheap and affordable technology, particularly a tablet-based kiosk platform with Bluetooth thermal printing, the system will be able to improve the working efficiency of the government while making accessibility easier to all residents, especially those who have limited access to digital resources, and promote transparency within local governance.

The following are the specific objectives of the study:

1. To develop a kiosk interface that allows residents to safely and easily request barangay documents such as Certificates of Indigency, Barangay Clearances, and other essential services with or without the need for a personal computer or smartphone.
2. To construct a secure and centralized database for storing and managing resident profiles, service requests, and other barangay-related records.
3. To create a web-based administrative dashboard accessible remotely by authorized barangay personnel for real-time management, tracking, and processing of requests, including adding, editing, and searching residents.
4. To automate the printing of a receipt for the walk-in and provision of SMS notifications to alert residents of document availability for collection.
5. To develop offline-first functionality that guarantees continuous system operation during internet outages and allows automatic data synchronization once connectivity is restored.
6. To implement basic user authentication and security measures that help ensure data privacy and prevent unauthorized access.
7. To test and evaluate the system's usability, reliability, and efficiency through deployment in one or two pilot barangays.

1.5. Scope and Delimitations.

This study mainly covers the design, development, and evaluation of the prototype for the Smart Barangay Kiosk System. The scope of this study is limited to the following:

- Focuses on designing, developing, and evaluating a Smart Barangay Kiosk System using a tablet kiosk with Bluetooth thermal printer.
- Focuses on designing, developing, and evaluating a Smart Barangay Kiosk System using a tablet kiosk with Bluetooth thermal printer.
- Testing and implementation are limited to Barangay Maangas and Barangay Sta. Maria in Presentacion, Camarines Sur.
- Payment is cash only upon pickup.
- The prototype will support requests for common barangay documents such as the following:
    - **Barangay Clearance** – is a Philippine government document from your local village (barangay) confirming you're a resident with good moral standing, with no pending issues, needed for jobs, business permits, loans, or opening bank accounts.
    - **Certificate of Indigency** – an official document issued by the barangay certifying that a resident is a member of an indigent or less fortunate family.
    - **Certificate of Residency** – an official document verifying that an individual is a legal and bona fide resident of a particular barangay.
    - **Certificate of Good Moral Character** – a document certifying that the resident is of good moral character as attested by the barangay.
    - **Certification for First Time Job Seeker** – a document from the barangay hall, issued under Republic Act 11261 (First Time Jobseekers Assistance Act), that certifies the resident is seeking first-time employment.

- **Certification for Monthly Income** – a document certifying the estimated monthly income of a resident, often required for scholarship applications and government aid programs.
- **Certification for Solo Parent** – an official document issued by the barangay, in coordination with the DSWD, that certifies an individual's status as a solo parent under the law.
- **Certification PDA** – a certification for Persons Deprived of Assistance, verifying a resident's eligibility status.
- **Business Clearance** – a clearance document required for business permit applications and renewals within the barangay.
- **Barangay ID** – an identification card issued by the barangay to verified residents.
- **Barangay Certification** – a general-purpose certification document issued by the barangay for various official purposes.
- **Other documents** – a custom option allowing residents to specify a document type not listed above.

The following are the delimitations of the study:

- No integration into any national government system nor any third-party platform.
- No mobile app development except for the Progressive Web App (PWA) version.
- Security is just basic, advanced cybersecurity is not being studied.
- Evaluation is limited to 2 pilot barangays, which may not represent the entire barangay.

CHAPTER 2

RELATED LITERATURE REVIEW AND STUDIES

2.1. Theoretical Basis

This section describes theories which form the core of the design of the Smart Barangay Kiosk System; e-governance, offline-first web architectures, and the principles of human-centered design. These explain in a logical and scientific manner why we were able to develop a low-cost yet highly resilient system that responds to specifics of barangay-level issues in the Philippines such as erratic connectivity and low technical literacy among the population.

2.1.1. E-Governance

E-Governance refers to the employment of information and communication technologies (ICT) meant to improve the delivery of public services and transparency and citizen engagement in governmental processes. This, according to the DILG, further enhances the PDP 2023-2028 through accountability and efficiency in local administrations in barangay-level e-Governance. Agents like the Seal of Good Local Governance for Barangays (SGLGB) and the Retooled Community Support Program (RCSP) demand that digital means be explored for lessening the bureaucratic bottleneck in document requests and resident management systems, thus contributing to the national goal of developing truly inclusive communities [9]. The Smart Barangay Kiosk System further applies this through provisions made for offline document requests, complemented with SMS alerts, resulting in reduced queues and paperwork and ensuring traceability with numbers—the very reason for DILG promoting digitized barangay services in low-resource settings.

2.1.2. Offline-First Progressive Web Apps (PWAs)

Offline-first PWAs encourage online access and service-workers can be used for local storage and caching operations (e.g., IndexedDB), so that synchronization can automatically be resumed once the app comes back online. The architecture is essentially one of progressive enhancement, where core functionalities are targeted for offline use, and enhancements depending on network availability are added for online use [10]. In the Philippine context, where connectivity compromises reliability for rural barangays, they are enabling the development of resilient apps such as the Power Kiosk System. With Supabase employing local cache for offline handling of requests, action is synchronized in real-time, thus assuring complete offline work for residents' submissions and approvals of the administration during the operational phase of the site [11].

2.1.3. HCI for Low-Literacy Users

HCI states that interfaces should seek to be intuitive and minimize cognitive load, especially for a broad range of users such as elderly residents or with limited exposure to technology. A recent study pertaining to rural UX design in the Philippines has advocated multilanguage support (English/Filipino) and a large-button cutout touch interface for maximum accessibility [12]. In the context of the current work, the simple nature of the kiosk forms (name, address, purpose) with receipt printing automatically minimize errors, and the responsive design of the PWA shall be undercut on phones without any login hurdles posed to the residents, thereby increasing usability of the system in barangay halls in case of brownouts and long queues.

2.1.4. Database Management & BaaS

Modern DBMS today have the full capability of ensuring data integrity in real-time while synchronizing offline; for example, Supabase (PostgreSQL-Based BaaS) provides this feature through edge functions with local caching. This is for the Philippine capstone projects for the years 2023-2024 since Supabase would be the most appropriate choice for the scalable yet cheap backends of community apps of which these would have authentication and SMS integrations without the burdens of dealing with heavy server management. In this instance, Supabase is acting on the level of resident records and pending requests in auto-sync mode, thereby maintaining consistency between kiosk and admin dashboard while being compliant with the data privacy restrictions laid out in the Data Privacy Act of 2012 [13].

2.1.5. Tablet-Based Kiosk Deployment

The use of consumer-grade tablets as kiosk platforms has become an increasingly viable alternative to custom embedded hardware. Android tablets provide touchscreen access together with Wi-Fi and Bluetooth wireless connections and they include built-in battery power and the capability to operate Progressive Web Apps through the Chromium browser at a much lower price than dedicated kiosk equipment costs [7]. The tablet kiosk system achieves receipt printing functionality which matches the capabilities of standard POS systems through its connection to Bluetooth thermal printers like the GOOJPRT PT-210 while remaining lightweight and energy-efficient. The barangay hall space requirements together with their budget constraints and their limited technical knowledge make this system design most appropriate for their facilities.

2.1.6. System Security and SMS Integration

All security provisions with respect to e-Governance systems have been taken care of regarding authentication, encryption, and role-based access to sensitive resident data. The Philippine guidelines from DILG want secure APIs and hashing for barangay archives. The study employs Supabase Auth with bcrypt password hashing, rate-limited kiosk password validation, OTP-based password recovery via PhilSMS, and comprehensive audit logging for all administrative sessions. SMS notifications are sent through the PhilSMS gateway for affordable and reliable notifications in the Philippines when documents are ready for pickup [11].

2.2. Related Studies

This section considers several studies, both foreign and local, about barangay management and kiosks as innovations and gaps. While some prior works did digitization of records, very few integrated full offline PWAs, physical kiosks, or SMS in real deployments-these gaps were addressed by this study.

2.2.1. Community Data Management Using IoT Devices (Foreign)

In the year 2019, Adekunle and Oladipo developed an IoT-enabled kiosk for rural communities in Africa for the self-service submission of forms to be kept locally, with auto-sync. The multilingual interface, aware that the target users may have low literacy skills, emphasized maximum robustness in a low-connectivity environment to limit manual bottlenecks [16].This validates offline kiosks for developing economies, but they do not include SMS or integration with Philippine LGUs for alerts on the readiness of documents, which is integrated into Smart Barangay Kiosk.

2.2.2. Design and Implementation of a Local E-Government Portal (Local)

Reyes and Salonga (2021) have established the online portal for barangay clearance services. They created a section for people that serves as a tracking status and notification in real-time. Although this improves access, not all areas have a constant internet connection, rendering it void. Usability evaluation reduced foot traffic; however, the results showed that hybrid offline systems are necessary. This study builds upon it by using PWA offline caching and kiosk hardware [17].

2.2.3. Barangay Management Information System (Local)

According to Hernandez and Cruz (2021), resident records and permits were digitized and files were created to streamline administering tasks and reduce errors through the use of SQL databases [18]. However, it lacked functionality for self-service by residents or offline modes, thus requiring visitations to the office. This system tends to extend that by allowing no-login requests and thermal printing.

2.2.4. WebYu: A Barangay Yuson Web-based Information Management System (Local)

Santos et al. in 2025 deployed WebYu in Barangay Yuson, Nueva Ecija, using Scrum methodology for automated workflows and with components of user experience (UX) evaluated using Technology Acceptance Model (TAM) having a mean score of 4.74 out of 5. It decreased certification to only 32% of its former duration, but it relied on online access. The idea of the dashboard, therefore, is to enhance resilience through offline PWA and SMS [19].

2.2.5. Development of an Offline-Capable Barangay Health Management System (Local)

In 2024 Lim and Torres had created a mobile PWA for barangay health records, with offline synchronization using Firebase, which could reduce 50% of data entry time across two pilot sites, for health data tracking alone, and meant to work in areas where there is low bandwidth, excluding general requests for documents or printing [20]. This credit goes towards PWA-BaaS hybrids for barangays, and kiosk is an add-on to multi-document workflows using hardware intergration [20].

2.2.6. E-Barangay Portal with SMS Integration for Resident Services (Local)

Dela Cruz et al. (2023) developed an online portal for clearances with Twilio SMS notifications resulting in 40% better response rates in urban barangays. Yet, this resulted in a 25% failure rate in their offline use because of outages. Thus, this upholds the SMS utility, which the kiosk-combined full offline PWA offers to ensure there is coverage [22].

2.2.7. Enhanced Barangay Information Management System with Residency Certificate Issuance (Local)

Imus and colleagues (2024) designed a web-based system for the issuance of certificates of residency, which was evaluated via USE questionnaire among officials and IT practitioners (mean score: 4.58/5) [23]. It had the requisite infrastructure for Web-based issuance and no offline kiosk hardware or other document support. This study extends the previous effort by incorporating PWA offline synchronization and thermal auto-printing for walk-ins[23].

2.2.8. One Barangay: A Mobile and Web Barangay Management System (Local)

According to Mirasol et al. (2023), they created a hybrid mobile-web system to centralize record keeping and avoid corrupt data caused by manual Excel usage and tested in urban settings: besides improved security, one disappointment was the need for persistent connectivity, excluding offline conditions prevalent in rural settings [24]. The Smart Barangay Kiosk addresses this by providing Supabase local cache and IndexedDB durability for rural areas.

2.2.9. Barangay Certificate Issuance and Record Management System with Descriptive Analytics (Local)

Navarro et al. (2025) developed a web application using PHP/MySQL for the analytics of blotter and population reports that enable sorting data by gender/age. While this provided a better understanding for the admin, it lacked the development of self-service kiosks or SMS [25].This informs dashboard analytics extended here with offline PWA requests and real-time notifications.

2.2.10. A Risk Assessment of a Barangay Management System Portal (Local)

The studies made by Fernando et al. (2024) have shown that among the web portals geared towards developmental programs, ICT deficiencies and risks from manual operation pointed out were within Philippine barangays [26].Data handling vulnerabilities are mentioned, but hardware solutions are not recommended. At present, the Pi-based kiosk counter soaks up most of this mess with encrypted offline storage and role-based access [26].

2.3. Synthesis and Research Gap

The research shows Philippine e-Governance development through three stages starting from basic digital transformation according to Hernandez and Cruz 2021 and Reyes and Salonga 2021 through its current stage which uses limited offline capabilities according to Lim and Torres 2024 and Santos et al 2025 while the system uses international kiosk design according to Adekunle and Oladipo 2019. The new studies by Imus et al 2024 and Navarro et al 2025 concentrated more on analyzing user experience through usability testing and data analysis methods. The system currently lacks an integrated solution which enables users to undergo offline self-service operations through progressive web applications and physical kiosks while using thermal printers and SMS notifications to print multiple documents in true multi-barangay situations.

The research establishes these two barangays as testing sites for Supabase-powered PWA operations through a tablet kiosk which uses Bluetooth thermal printer technology to enable cash-only pickups and administrative functions. The research results support national e-Government objectives because they show potential solutions which can help rural areas by providing scalable tools and solutions which automatically synchronize data in areas with poor internet access.

CHAPTER 3

METHODOLOGY

3.1. Research Design and Systems Model

The study followed the Design and Development Research (DDR) method [29], in particular, the prototype development model for the innovation of the Smart Barangay Kiosk System with Offline-First Document Request Handling and Resident Management. DDR was appropriate for the study as it involved the design, development, and evaluation of a technology-based solution aimed at improving barangay services through digital transformation, with a focus on offline capabilities to address unreliable internet access in rural areas.

The process included planning, system design, prototyping, testing, and evaluation. The study will also follow an Iterative Development Life Cycle model consisting of the following phases: Requirements Analysis, System Design, Development and Implementation, Testing and Evaluation, and Refinement and Finalization.

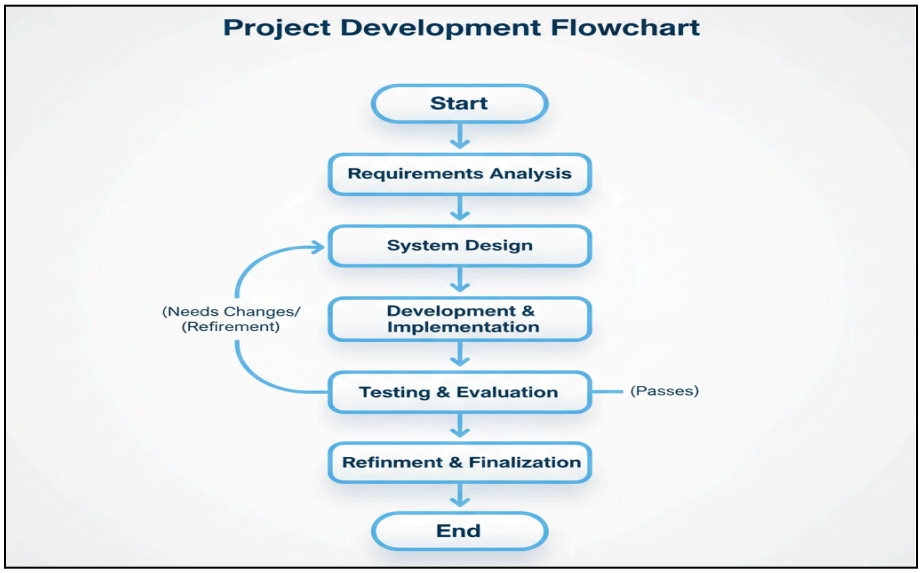


Figure 3.1: Conceptual Framework of the Study

Each phase includes the following: For the Requirement Analysis phase, it involves identifying the project's requirements and objectives. The Data Collection Method gathers data through interviews and surveys, observation, and document analysis that help the researchers identify the problems to be addressed.

The System Design, Development, and Implementation phase builds the system using specific development tools and technologies. This phase encompasses the kiosk creation and development process involving hardware setup, software configuration, interface and application development, and thorough kiosk testing.

The system features based on objectives are evaluated during the Testing and Evaluation phase. Target respondents and sampling involve identifying participants, such as barangay officials and residents, and determining the sampling technique.

In addition, the model includes Development Boundaries and Prototype Constraints as part of Refinement and Finalization. Finally, Ethical Considerations are dealt with to ensure the accountability and integrity of the entire project.

3.2. Requirement Analysis and Data Collection Methods

To guarantee that the Smart Barangay Kiosk System provides actual services and addresses the real needs and challenges of barangay service delivery, multiple data collection techniques were employed. These were selected to be the most relevant qualitative and quantitative data to inform the design, development, and functionality of the system.

3.2.1. Interviews and Surveys

The researchers conducted structured interviews and guided surveys with selected barangay officials and residents from Barangay Maangas and Barangay Sta. Maria in Presentacion, Camarines Sur. The study aimed to assess current service delivery methods from both administrative and user experience perspectives. The barangay officials, which included the secretary and clerks and the barangay captain, shared their experiences about document processing delays and data storage challenges and communication problems with residents and their requirement for both offline functions and SMS notification services.

Similarly, residents from diverse backgrounds—including senior citizens, persons with disabilities (PWDs), and low-income individuals—were surveyed to determine their level of access to digital tools, common service needs, and barriers encountered when availing of barangay services. The survey used closed questions together with open-ended questions to collect numerical data about service usage and waiting periods and to gather respondent feedback about their satisfaction and improvement recommendations.

The system's survey module enabled the creation and distribution of four mapped survey tools which utilized a 5-point Likert scale rating system (1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree). The instruments and their items are as follows: The instruments and their items are as follows:

A. Resident Pre-Usage Survey (15 items)

Administered before the resident uses the kiosk system, covering four parts:

- **Part 1 – Current Experience with Document Requests (5 items):**

1. I am satisfied with the current manual process of requesting barangay documents.
2. The current process of requesting barangay documents is fast and efficient.
3. I find it convenient to personally go to the barangay hall every time I need a document.
4. I rarely experience long waiting times when requesting documents at the barangay.
5. I am always informed about the status of my document request (e.g., if it is ready for pickup).

- **Part 2 – Accessibility and Inclusivity (3 items):**

6. The barangay's current document request process is accessible to all residents, including senior citizens and PWDs.
7. I can easily request barangay documents even during my work hours or busy schedule.
8. I have no difficulty communicating my document request needs to barangay staff.

- **Part 3 – Technology Readiness (4 items):**

9. I am comfortable using digital devices such as smartphones, tablets, or computers.
10. I am familiar with using self-service kiosks or touch-screen machines (e.g., ATMs, payment terminals).

11. I have access to a smartphone or computer with internet connection at home.
12. I am open to using a computerized system for requesting barangay documents.

- **Part 4 – Perceived Need for Digital Innovation (3 items):**
    13. I believe the barangay needs a digital/computerized system for processing document requests.
    14. A system that allows me to request documents from home (online) would be very helpful.
    15. Receiving SMS notifications about my document request status would improve my experience.

B. Resident Post-Usage Survey (23 items)

Administered after the resident has used the kiosk system, covering six parts:

- **Part 1 – Ease of Use / Usability (5 items):**
    1. The system was easy to understand and navigate, even without prior training.
    2. The on-screen instructions and labels were clear and easy to follow.
    3. I was able to complete my document request without needing assistance from barangay staff.
    4. The system's interface design is user-friendly and visually appealing.

5. I feel confident that I can use the system again on my own in the future.

- **Part 2 – Efficiency and Speed (3 items):**
    6. Using the system, I was able to request a document faster compared to the traditional manual process.
    7. The number of steps required to complete a request was reasonable and not excessive.
    8. The system processed my request without significant delays or errors.
- **Part 3 – Accessibility and Convenience (4 items):**
    9. The kiosk at the barangay hall is easy to access and use for walk-in residents.
    10. The option to request documents online (remotely) is very convenient.
    11. The system makes barangay document requests more accessible to senior citizens, PWDs, and busy residents.
    12. I can request documents at a time that is convenient for me, not limited to office hours.
- **Part 4 – Reliability and Offline Capability (3 items):**
    13. The system worked properly even when internet connection was slow or unavailable.
    14. I did not experience any system crash or major technical issue during my use.

15. I trust that my document request was properly recorded and will be processed.

- **Part 5 – SMS Notification and Communication (3 items):**
    16. I received an SMS notification about the status of my document request.
    17. The SMS notification was timely and helped me know when to pick up my document.
    18. The SMS feature reduced the need for me to follow up in person at the barangay hall.
- **Part 6 – Overall Satisfaction and Impact (5 items):**
    19. Overall, I am satisfied with my experience using the Smart Barangay Kiosk System.
    20. The system significantly improved the way I request barangay documents compared to the old process.
    21. I would recommend this system to other residents in our barangay.
    22. I prefer using this system over the traditional manual process for future document requests.
    23. I believe this system should be permanently adopted by the barangay.

C. Admin Pre-Usage Survey (14 items)

Administered to barangay officials and staff before using the admin dashboard, covering four parts:

- **Part 1 – Current Document Processing Workflow (4 items):**

1. I am satisfied with the current manual process of handling barangay document requests.
2. The current manual process allows me to process document requests quickly and efficiently.
3. It is easy to keep track of all pending, in-progress, and completed document requests manually.
4. I rarely encounter problems with lost or misplaced document requests in the current system.

- **Part 2 – Challenges in Current Operations (4 items):**

5. Managing document requests during peak hours is difficult with the current manual process.
6. Residents frequently follow up in person about the status of their document requests.
7. Record-keeping and generating reports on document requests is time-consuming.
8. I sometimes have difficulty verifying resident information for document processing.

- **Part 3 – Technology Readiness (3 items):**

9. I am comfortable using computers and digital tools for my work at the barangay.
10. I am willing to learn and adapt to a new computerized system for managing document requests.

11. I believe that a digital system can improve the efficiency of our barangay services.

- **Part 4 – Expectations for Digital Innovation (3 items):**

12. I expect a kiosk system to reduce the workload of barangay staff in processing documents.
13. A system that allows online/remote document requests would greatly help our barangay operations.
14. Automated SMS notifications for residents would reduce follow-up inquiries to barangay staff.

D. Admin Post-Usage Survey (19 items)

Administered to barangay officials and staff after using the admin dashboard, covering five parts:

- **Part 1 – Ease of Use / Usability (4 items):**

1. The admin dashboard was easy to understand and navigate, even without extensive training.
2. The interface for managing document requests (view, process, release) is clear and intuitive.
3. I can easily track the status of all requests from pending to released.
4. The overall layout and design of the admin panel is user-friendly and visually organized.

- **Part 2 – Efficiency and Speed (4 items):**

5. Using the system, I can process document requests faster compared to the manual process.
6. The request queue system (pending → in progress → ready → released) improved my workflow.
7. The system reduced the number of in-person follow-ups from residents about their request status.
8. Record-keeping and viewing release logs is easier with the system than with manual records.

- **Part 3 – System Features (4 items):**
  9. The SMS notification feature is useful for informing residents about their document status.
  10. The resident verification feature helps ensure only legitimate residents can request documents.
  11. The calendar and announcements features are useful for managing barangay communications.
  12. The feedback from residents displayed in the system helps improve barangay services.
- **Part 4 – Reliability and Offline Capability (3 items):**
  13. The system worked reliably even during slow or intermittent internet connections.
  14. I did not experience system crashes or data loss while processing requests.

15. I trust that all document requests and release logs are properly saved in the system.

- **Part 5 – Overall Satisfaction and Impact (4 items):**

16. Overall, I am satisfied with the Smart Barangay Kiosk System as a tool for managing document requests.
17. The system has significantly improved the efficiency of our barangay document processing.
18. I would recommend this system to other barangays for managing their document request services.
19. I prefer using this system over the traditional manual process for future document management.

In addition to the Likert-scale instruments, the system also includes a built-in feedback module that prompts users to rate their experience on a 1–5 star scale and provide optional free-text comments after a document request is submitted via the kiosk, processed and released by the admin, or manually encoded by the admin.

3.2.2 Observation

The service delivery system for Barangay Maangas and Barangay Sta. Maria in Presentacion, Camarines Sur underwent evaluation through direct observation methods which do not need any invasive procedures. The research team visited barangay halls during regular office hours to observe how documents are requested and processed, how staff members interact with residents, how announcements go out to the public, and how records will be kept and preserved.

The following key observations were noted during the site visits:

1) **Manual, paper-based workflow:** Both barangay halls relied entirely on handwritten forms and logbooks for recording document requests. Residents were required to fill out a physical request form, after which the secretary or acting secretary manually typed the document using a word processor or pre-formatted template. This process was time-consuming and prone to typographical errors.
2) **No centralized record-keeping:** The organization kept its document request logs through physical ledger systems and used basic spreadsheet files that operated on one computer system. The absence of a unified digital database system created challenges for users to access past records and monitor ongoing requests and produce summary documents.
3) **In-person follow-ups:** The residents lacked any method to check their document request status from a distance. The residents returned to the barangay hall for follow-up visits because they needed updates on their document requests which created traffic during busy time periods. The admin pre-usage survey showed that residents typically visited the office to check their document status which received a 4.22 average rating.
4) **Limited office hours and accessibility:** Barangay services were available only during regular office hours, creating difficulty for working residents. Senior citizens and persons with disabilities (PWDs) also had to personally visit the barangay hall, as no remote or self-service option was available. The resident pre-usage survey confirmed this, with "I can easily request documents during work hours" receiving the lowest accessibility rating (M = 2.33).

5) **Single point of failure:** Document processing in both barangays relied on the work of either the barangay secretary or the appointed acting secretary. Document processing stopped because the person responsible was either not present or occupied with other tasks. The organization required a system that would handle requests from users without depending on staff members for assistance according to this observation.
6) **Announcement dissemination:** The barangay shared official announcements through two main channels which included physical bulletin boards and Facebook groups. The announcement system required direct delivery to residents from the service point but it lacked an integrated mechanism.
7) **Intermittent internet connectivity:** The two barangay halls encountered repeated internet outages which proved the requirement for an offline-first system that enables document request processing and storage of requests into queueing for later times without needing internet access.

The system design decisions were based on these observations which specifically affected the self-service kiosk interface design (which solved items 1 and 5) and the centralized Supabase database (which solved item 2) and SMS notifications (which solved item 3) and the remote resident portal (which solved item 4) and the events/announcements module (which solved item 6) and the offline-first architecture using IndexedDB and Workbox service workers (which solved item 7).

3.2.3. Document Analysis

The team conducted a complete document analysis to test the kiosk system's ability to work with actual barangay administrative processes. They reviewed all existing documents and forms used by the barangay which included the standard certificates plus the feedback and complaint forms and the document request logs and their response time records and the formats used for announcements and the manual record systems and the Excel sheets and logbooks used for filing.

The document analysis was complemented by structured survey data gathered from a total of 9 barangay officials and staff—specifically the barangay secretaries and acting secretaries who are directly responsible for document processing. Five (5) respondents came from Barangay Maangas and four (4) from Barangay Sta. Maria, Presentacion, Camarines Sur. Each respondent completed both a pre-usage survey (14 items) and a post-usage survey (19 items) through the system's built-in survey module.

The following findings emerged from the document analysis and the administrative survey data:

1. **Document templates and formatting:** Physical document templates were collected from both barangays. The formats varied slightly between the two (e.g., header layouts, signatory positions, official seal placement), confirming the need for a configurable, per-barangay template system rather than a one-size-fits-all approach. The system's PDF generation module was designed with this flexibility, supporting customizable barangay name, officials, signatures, seal images, and pricing per document type.

2. **Volume and types of requests:** Both barangays reported that Barangay Certification, Certificate of Indigency, and Barangay Clearance are the three most frequently requested documents. This was validated during the pilot deployment, where these three types accounted for 76.7% of the 86 documents released (32 Barangay Certifications, 20 Certificates of Indigency, and 14 Barangay Clearances).
3. **Record-keeping challenges:** The administrative pre-usage survey results showed that record-keeping and report generation work took the longest time under the current system because it received the highest challenge score of 4.78 with a standard deviation of 0.44. The physical logbook entries showed irregular patterns because they contained blank spaces and missing date entries which made it impossible to track previous records. This situation demonstrated the essential requirement for a digital database system which would enable users to search through complete release documentation.
4. **Verification difficulties:** The secretaries encountered problems when they tried to confirm resident information needed for document processing which resulted in an average score of 4.56 and standard deviation of 0.73 on the admin pre-usage survey. The current procedure depended on secretaries to use their personal knowledge about residents or to perform manual checks of registration logbooks. The system developed its resident database together with its verification module which offers a searchable

resident registry containing personal details and zone information and account status.

5. **Workflow inefficiency during peak hours:** The admin pre-usage survey showed strong agreement (M = 4.44, SD = 0.53) that managing requests during peak hours is difficult under the current manual process. The system implemented a digital request queue with status tracking capabilities which allowed multiple requests to be accepted through the kiosk while the secretary continued to handle prior request work.
6. **Post-deployment improvements:** The nine system admin respondents who evaluated the system after its usage assigned a rating of 4.52 out of 5.00 for the complete set of 19 post-usage evaluation items. The record-keeping process and release log viewing function achieved a post-usage score of 4.67 while the manual record-keeping pre-usage challenge received a score of 4.78, which shows that the system effectively solved the specific problem that had been identified. All admin respondents showed their system satisfaction with a mean score of 4.72 while all nine respondents confirmed their desire to recommend the system to other barangays with a mean score of 4.78.

3.3. System Design, Development and Implementation Tools and Technologies

The Smart Barangay Kiosk System Development used cost-effective low-power hardware and open-source software solutions to achieve budget-friendly operational costs and system expansion and long-term system maintenance. The project selected tools and

technologies which fulfilled its objectives by providing offline functionality and simple installation and user access in areas with limited resources.

3.3.1. Hardware Components

1. **Android Tablet:** The main kiosk system uses an Android tablet that people can buy at retail stores. The device uses a capacitive touchscreen display as its user interface element while providing Wi-Fi and Bluetooth connectivity and built-in battery power for brownout protection and full-screen PWA operation through the Chromium browser. Residents visiting the barangay hall can use the tablet system which the tablet maintains at that location.
2. **Bluetooth Thermal Printer (GOOJPRT PT-210):** The Bluetooth thermal printer which has a 58mm size operates with minimal maintenance to print receipts automatically for customers who request documents in person. The printer establishes a Bluetooth Classic connection through the RawBT Android application to verify receipt of control numbers and document lists and pricing details from the tablet.

Figure 3.2: Image of the physical kiosk setup (tablet + Bluetooth printer)

3.3.2. Software and Development Tools

1. **Frontend Technologies (React + Vite):** The user interface of the kiosk was developed as a Progressive Web App (PWA) using React for component-based structure and Vite for fast development and bundling. This enables a single app that works seamlessly on kiosks and phone browsers, with offline support via service workers.
2. **Backend and Database (Supabase):** Supabase (the free tier) handled backend services, authentication, and database management. It provides offline caching, real-time updating, and automatic syncing upon returning to the internet, thus guaranteeing for fully offline operation of document requests and resident management.

3. **Offline Sync (IndexedDB + Workbox):** The custom offline storage module used IndexedDB (database name: sbk-offline) with four object stores: barangays, residents, settings, and pendingRequests. The settings store functions as a general-purpose cache which stores all resident portal data using prefixed keys for session credentials and profile and resident record and document options and pricing and announcements and zone counts and request history. Workbox runtime caching implements NetworkFirst strategy for Supabase REST API calls which have a 5-second timeout and a 200-entry limit and a 24-hour expiry period while using CacheFirst strategy for Supabase Storage assets. The useOfflineSync hook enables automatic request flushing from pendingRequests store to Supabase when internet connection becomes available again.
4. **Thermal Printing (RawBT + ESC/POS):** The RawBT Android application enabled receipt printing by connecting the PWA to Bluetooth thermal printers through Android intent URL scheme (rawbt:{encoded_text}) technology. The receipt content extends to 32 characters per line on 58mm paper width which displays barangay name and date and reference number and queue number and resident name and document list and pricing details and a footer message.
5. **SMS Notification (PhilSMS):** The system used PhilSMS to deliver cost-effective SMS alerts that inform users about document pickup times in the Philippines. The system also supports Semaphore as an alternative provider. The SMS edge functions perform phone number normalization which follows the PH 63 format and generates one-time passwords for password recovery through 6-digit codes that expire after 5 minutes and sends document-ready alerts.

6. **Hosting:** The PWA was hosted on Vercel for the frontend. Supabase free tier handles the backend, database, and edge functions. Dynamic manifest switching allows separate PWA installs for kiosk mode (/kiosk), admin mode (/admin), and super-admin mode (/superadmin).
7. **Security Measures:**
    - *Authentication:* The system uses Supabase Auth with email/password authentication for both admins and residents while storing separate bcrypt-hashed kiosk passwords for each barangay in `app_settings`
    - *Access Control:* The system uses role-based access control (superadmin, barangay_admin, resident) together with RLS policies to enforce security restrictions. The `current_admin_barangay()` and `is_superadmin()` PostgreSQL functions enforce tenant isolation according to their design.
    - *Rate Limiting:* The system permits a maximum of 5 kiosk password validation attempts which must be completed within a 15-minute time frame.
    - *OTP Password Recovery:* SMS-based OTP codes with 5-minute expiry and attempt tracking.
    - *Audit Logging:* The `audit_logs` table records all superadmin and admin activities which include actor information, action details, and target data together with additional metadata.

- *Session Management:* Automatic idle reset after 2 minutes on the kiosk interface; lazy Supabase client creation per route to prevent token refresh issues.

3.3.3. System Architecture and Module Design

The system was architected as a multi-tenant PWA with four primary interfaces:

1. **Kiosk Interface (/kiosk):** The public-facing self-service terminal deployed on the tablet. Residents conducted name searches which lead them to choose document types and complete required fields for purpose and CTC number for business clearances before they submit their information. The system creates a control number which produces a thermal receipt and places the request into the administration processing queue. The system saves requests to IndexedDB when it operates offline and it will later synchronize the data once the system regains network access.
2. **Resident Portal (/resident):** A login-protected interface for verified residents to submit document requests remotely from their phones, view request history with status tracking, submit feedback and ratings on released documents, and manage their profile information. After a resident's account is verified at least once while online, the portal caches session credentials, profile data, barangay information, document options, pricing, announcements, and request history in IndexedDB. Subsequent logins can proceed entirely offline using the cached session, and document requests submitted offline are queued locally with an offline reference number and automatically uploaded when internet connectivity returns. An offline status banner informs the user of connectivity state and pending request count.

3. **Admin Dashboard (/admin):** A comprehensive management panel with nine tabs:
    - *Dashboard:* Daily summary with key metrics (new submissions, pending verifications, announcements, pricing gaps), officials status, and calendar events.
    - *Manage Residents:* CRUD operations for residents with bulk import, search, and zone assignment.
    - *Requests:* Document request workflow (Pending → In Progress → Ready for Pickup) with SMS alerts, payment logging, release confirmation, and chat panel for internal notes.
    - *Verification:* Review and approve/reject resident registration submissions.
    - *Calendar:* Admin-created events with holiday integration.
    - *Announcements:* Create/edit barangay announcements with image uploads and date ranges.
    - *Pricing:* Configure document service fees per barangay.
    - *Feedback:* View resident ratings (1–5 stars) and comments with CSV export.
    - *Barangay Info:* Manage officials, zone settings, feature toggles, and seal upload.
4. **Super-Admin Panel (/superadmin):** The system provides a management interface which includes thirteen tabs for users to access various functions including overview displays and analytics tools and broadcast announcement systems and tenant management tools which allow create and edit and delete

functions for barangays and admin users can manage system users and the system enables viewing of resident data and request information while document configuration and audit log and feedback aggregation and data cleanup utilities and access and security settings and thesis document generation functionalities are available.

3.3.4. System Flowchart

Figure 3.3 illustrates the complete system flowchart of the Smart Barangay Kiosk System, showing the four user entry points and their respective workflows converging on the shared cloud backend. The system operates through four distinct interfaces, each serving a specific user role:

1. **Walk-in Kiosk Flow (Tablet at Barangay Hall):** The home screen shows barangay announcements to greet the resident. The kiosk checks whether the resident exists in the database via name search (querying the IndexedDB cache first, then Supabase if online). The system populates resident information automatically when the resident exists in the database; otherwise, the new applicant intake form collects resident demographic information. The resident chooses document(s) and provides purpose and CTC details when needed before checking the document pricing which includes the document cost and service fee. The system verifies connection status after submission: request transmission happens through POST to Supabase when the system detects online status; the system saves request data to the IndexedDB pendingRequests store for future automatic synchronization when the system regains connectivity.. A thermal receipt is printed via the RawBT Android app to the GOOJPRT PT-210 Bluetooth

printer. Finally, the resident is prompted for feedback (1–5 stars + optional comment) and a Likert-scale survey. The kiosk auto-resets after 2 minutes of idle time.

2. **Resident Portal Flow (Remote Web Access):** Remote residents authenticated via email and password through Supabase Auth. New users register and link their account to an existing resident record (by matching name, birthday, zone, and contact details) or register as a new resident. Once linked, residents can select documents, submit requests (source: resident_portal), view request history and release logs, provide feedback on released documents, and complete pre/post-usage surveys. The portal supports offline session caching via IndexedDB for continued access during connectivity interruptions.
3. **Admin Dashboard Flow (Secretary/Staff):** Barangay officials need to logged in using their email and password which the system verifies through the admin_users table. The dashboard shows a request pipeline which includes four different status levels that start from Pending (new submissions from kiosk and portal) to Current (being processed by staff) and then to Done (ready for pickup which triggers SMS notification via PhilSMS) before reaching Released (archived with payment recorded in release logs). The system provides extra management tabs which allow approval or rejection of new profiles for resident verification together with resident database management and announcement scheduling and document pricing configuration and barangay information setup and official creation and feedback evaluation. The system triggers admin surveys at two points which are activated during login and after the first document release.

4. **Super Admin Flow (System-wide Management):** The super admin used a unified panel to manage multiple tenants through health dashboards for each barangay and cross-barangay analytics that allow CSV exports and broadcast announcements and tenant creation with configuration and admin user management and audit log access and system-wide setting control.
    - Offline-First Synchronization: When someone submits a kiosk request while they have no internet access the system archives the request in the IndexedDB pendingRequests object store. The useOfflineSync React hook operated as a network status monitor which sends all pending requests to Supabase when it detects an active internet connection. The system deletes successfully synced requests from the local queue while it keeps failed requests until the next synchronization period. Workbox service workers deliver runtime caching support for Supabase REST API responses which use a NetworkFirst strategy that includes a 5-second timeout and a 200-entry restriction and a 24-hour response time limit and for Supabase Storage assets which use a CacheFirst strategy.
    - Cloud Backend: The four interfaces all connect to the Supabase cloud backend which includes a PostgreSQL database that implements Row-Level Security (RLS) policies to maintain multi-tenant data separation together with Supabase Auth for JWT-based user authentication at all user permission levels and Edge Functions which handle SMS sending (PhilSMS) and OTP creation and kiosk password checking and management of administrative users.

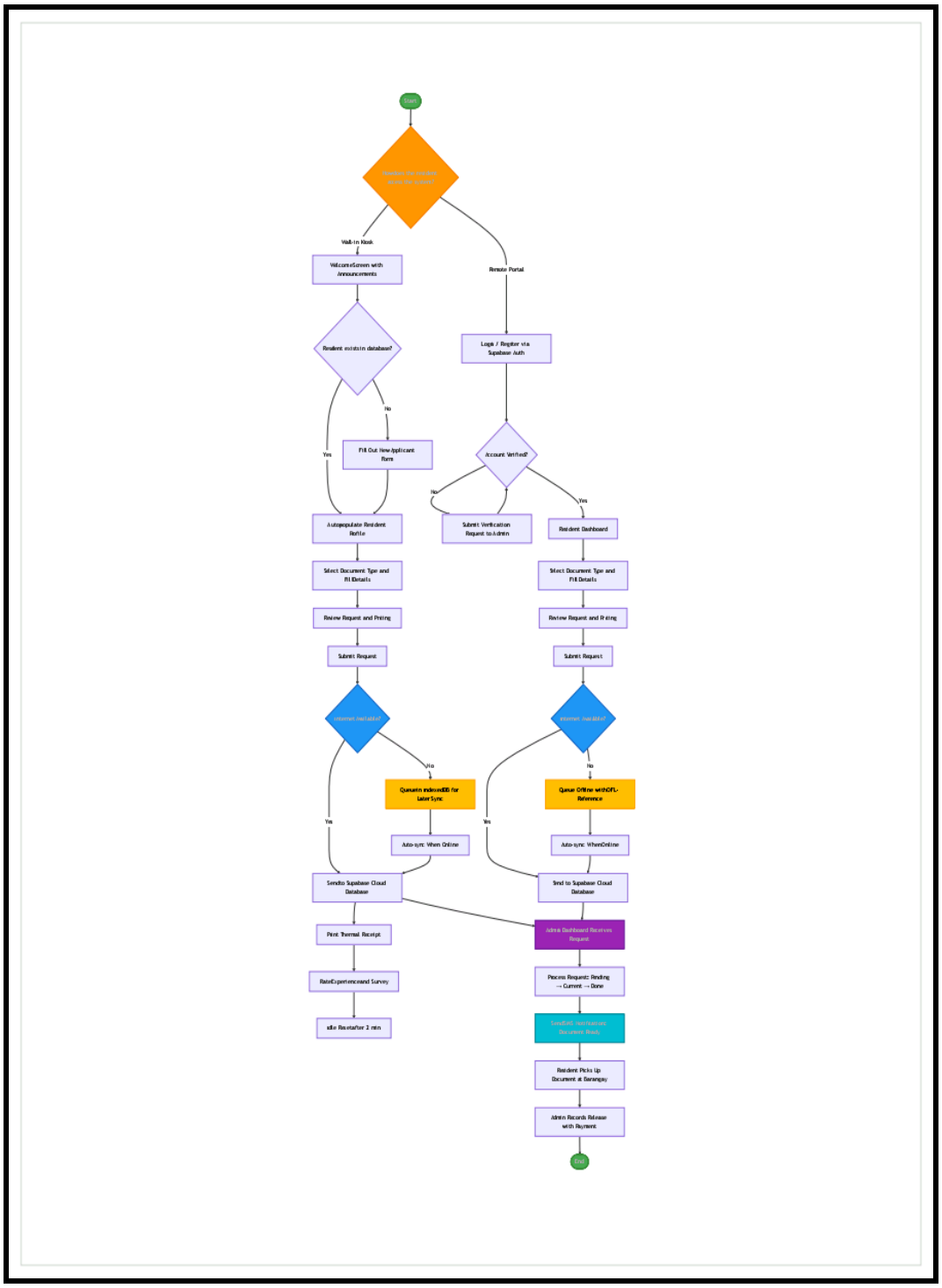


Figure 3.3. System Flowchart of the Smart Barangay Kiosk System with Offline-First Document Request Handling and Resident Management

3.3.5 Database Design

The system uses a PostgreSQL database managed by Supabase with 21 core tables organized around a multi-tenant architecture. All tenant-specific tables include a barangay_id foreign key with row-level security policies enforcing data isolation. The key tables are as follows:

Table	Purpose
`barangays`	Multi-tenant barangay registry with feature toggles, seal URL, and address information
`admin_users`	Admin accounts linked to Supabase auth with barangay assignment and role (barangay_admin, superadmin)
`residents`	Core resident demographic data (name, sex, birthday, civil status, address, occupation, education, religion)
`resident_profiles`	Extended profile data linked to Supabase auth for resident portal access (phone, email)
`resident_verification_requests`	Pending/approved/rejected new resident registrations with submitted data (JSONB) and admin notes
`resident_intake_requests`	Document requests with status workflow (pending/current/done/cancelled), queue number, and pricing
`release_logs`	Document release history with payment method, amount, and timestamps
`resident_feedback`	Star ratings (1–5) and comments per released document
`announcements`	Barangay announcements with image data, start/end dates

`admin_events`	Calendar events per barangay
`barangay_officials`	Officials registry (Punong Barangay, Kagawad, SK Chairperson, Treasurer, Secretary) with titles
`barangay_zone_settings`	Zone configuration per barangay with coordinates data
`app_settings`	Global and per-barangay configuration (kiosk password, service fees, SMS fees, document options)
`password_reset_otps`	OTP tracking with expiry, usage, and attempt counters
`audit_logs`	Superadmin/admin action tracking with actor, action, target, and JSONB metadata
`resident_accounts`	Resident portal account registry linking Supabase auth users to barangay tenants with account status (active/disabled)
`conversations`	Chat threads linked to document requests for admin–resident communication
`messages`	Individual messages within conversations with sender role (admin/resident/system) and read status tracking
`survey_responses`	Anonymous pre-usage and post-usage survey responses from kiosk, resident portal, and admin-entered surveys stored as JSONB
`kiosk_feedback`	Walk-in kiosk user satisfaction ratings and comments collected after document request submission

Table 3.1. Core Database Tables and Their Purposes

3.3.6. Offline-First PWA Behavior

The system was deployed as a Progressive Web App using React 19 + Vite + Supabase along with Workbox service workers. This enables three critical offline scenarios:

1. **Kiosk Mode (Tablet):** The kiosk runs the PWA in full-screen standalone mode on the tablet browser. The system functions 100 percent offline for all operations including form processing control number creation resident information retrieval and receipt printing. The system stores requests in IndexedDB for local access and begins automatic synchronization when the internet connection becomes available.The useOfflineSync hook monitors navigator.onLine state and flushes queued requests upon reconnection.
2. **Resident Portal Offline Access (Verified Accounts):** The system saves all essential information about the resident's session and profile together with their resident record and barangay list and document options and pricing configuration and announcements and zone settings and request history after the resident successfully completes their first online account verification. The resident can access their account through cached credentials to view their dashboard and announcements and previous requests while they are offline after their first internet visit. The system generates a temporary reference number beginning with OFL for offline requests which it stores in the pendingRequests IndexDB database. The system displays an offline status banner which shows both the current

connectivity status and the existing number of requests that need to be processed. The useOfflineSync hook automatically transmits all waiting requests to Supabase for uploading and updates the pending request count when the device establishes a new connection.

3. **Mobile PWA Install (Resident using own phone):** On the first visit with brief internet access, the browser prompts "Add to Home Screen." Once installed, the PWA becomes a standalone app on the phone. Residents can open the app anytime (even in airplane mode) to submit document requests. This combines the resident portal offline access described above with the PWA shell caching provided by Workbox service workers.

3.4. Project Planning and Execution Details

3.4.1. System Features Based on Objectives

Module	Description
Document Request Module	The system lets residents obtain documents through a walk-in kiosk which includes Barangay Clearance and Indigency and Residency and Good Moral and First Time Job Seeker and Monthly Income and Solo Parent and PDA and Business Clearance and Barangay ID and Barangay Certification and custom documents (auto-printed receipt) or online (downloadable/screenshot receipt). Generates instant control numbers and queue numbers.
Resident Management	Enables admin to add, edit, search, and bulk-import residents. Includes zone assignment and profile management.
Resident Verification	Allows admins to review, approve, or reject new resident registration submissions from the portal.
Notification System	Sends SMS notifications via PhilSMS every time a document is ready for collection, which would include pertinent details, such as control number and payment instructions.
Admin Dashboard	The dashboard provides real-time access through both phones and laptops

	which includes nine management tabs that enable complete barangay administration..
Super-Admin Panel	System-wide management with thirteen tabs for multi-barangay oversight, analytics, tenant management, and audit logging.
Resident Portal	The system provides a secure login interface which allows authenticated residents to request documents from remote locations and submit their feedback while tracking their document requests. The system enables complete offline functionality which requires initial user verification to allow users to access cached content and submit requests without internet connection and receive automatic content updates when they establish internet connection.
Offline Functionality & Sync	The kiosk and resident portal system function entirely offline through their use of IndexedDB technology and Workbox service workers. The kiosk stores resident details along with system settings and active requests while the resident portal saves session data and user profiles and barangay information and available document types and cost details and announcements and complete request records.Automatic Supabase synchronization occurs on reconnect via the useOfflineSync hook.
Announcements & Events	The system enables users to make barangay announcements which include visual content together with functions to manage calendar events through specific date range selection.
Feedback Collection	Star ratings (1–5) and comments per released document with CSV export for analysis.
Audit Logging	Comprehensive action tracking for all administrative operations with actor, timestamp, action type, target, and metadata.

Table 3.2. System Features Based on Objectives

3.4.2. Project Cost Estimation and Hardware Breakdown

The hardware components for each kiosk unit were selected to balance affordability, portability, and functionality. Two complete kiosk sets were assembled—one for each pilot barangay. Table 3.2a presents the per-unit and total hardware costs.

Component	Description	Qty	Unit Cost (₱)	Total (₱)
Android Tablet	Consumer-grade Android tablet with touchscreen, Wi-Fi, and Bluetooth	2	2,049.00	4,098.00
Bluetooth Thermal Printer	GOOJPRT PT-210 (58mm, portable)	2	979.5	1,959.00
Kiosk Casing	Custom blue box enclosure with wheels, paint, and assembly	2	1,122.00	2,244.00
			Total Hardware Cost	**₱ 8,301.00**
			Cost per Kiosk Unit	**₱ 4,150.00**

Table 3.2a. Hardware Cost Breakdown (2 Kiosk Units)

The hardware investment of ₱8,301.00 for two complete kiosk units which cost ₱4,150.00 each proves that tablet-based kiosks work better than standard PC systems. All software components used in the system are free or open-source, as detailed in Table 3.3.

Component	Technology Used	Cost
Frontend & PWA	React 19 + Vite + TailwindCSS	Free
Backend + Database + Hosting	Supabase (free tier)	Free
Frontend Hosting	Vercel	Free
Offline Cache & Sync	IndexedDB + Workbox Service Workers	Free
Bluetooth Printing	RawBT + GOOJPRT PT-210	Free (app)
SMS Notifications	PhilSMS	₱110.00 for 300 credits (~₱0.35+ per SMS)

Table 3.3. Software and Operational Cost Summary

3.4.3. Project Timeline and Development Schedule

The project was executed over a period of approximately four months, from January 2026 to April 2026. Table 3.3a summarizes the actual project timeline, activities, and deliverables for each month.

Month	Activities / Milestones	Deliverables
Month 1 (January 2026)	● Conducted interviews, surveys, observations, and document analysis in Barangay Maangas and Barangay Sta. Maria, Presentacion, Camarines Sur ● Finalization of system requirements with barangay officials (secretaries and acting secretaries) ● Collection and analysis of physical document templates from both barangays ● UI/UX design and wireframing of kiosk, admin dashboard, and resident portal interfaces ● Supabase project setup, database schema design, and multi-tenant architecture planning ● Domain registration and Vercel hosting configuration	● Data collection questionnaires and needs assessment documentation ● Approved system requirements ● UI/UX prototypes ● Supabase project with initial database schema and RLS policies
Month 2 (February 2026)	● Development of kiosk interface (React 19 + Vite PWA): welcome screen, resident search, new applicant intake, document selection, pricing calculation, and request submission ● Offline-first functionality using IndexedDB and Workbox service workers (caching of barangays, residents, settings, and pending request queue) ● Reference number and queue number generation ● Thermal receipt printing integration via RawBT Android app and ESC/POS formatting for GOOJPRT PT-210 ● Resident portal development: authentication (Supabase Auth), registration, profile linking, document request, and request history	● Working offline-capable kiosk module with receipt printing ● Functional resident portal with authentication and document requests

Month 3 (March 2026)	• Admin dashboard development: request pipeline (pending → current → done → released), resident management, verification workflow, announcements, pricing configuration, calendar, feedback viewer, and barangay info management • Super admin panel development: multi-tenant overview, analytics, broadcast, tenant management, admin user management, audit logs, data export, and data cleanup • SMS integration with PhilSMS for document-ready notifications and OTP-based password recovery • Hardware assembly: custom kiosk casing construction (blue box with wheels), tablet mounting, and Bluetooth printer pairing • Security implementation: role-based access control, bcrypt password hashing, 2-minute idle reset, rate limiting (5 attempts/15 min), and Row-Level Security policies • Full offline/online stress testing and synchronization verification • Deployment and staff training in Barangay Maangas (5 admin users) • Deployment and staff training in Barangay Sta. Maria (4 admin users)	• Complete admin dashboard with all management features • Super admin panel with multi-tenant capabilities • SMS notification system operational • Two fully assembled kiosk units • Live system deployed in both pilot barangays
Month 4 (April 1–17, 2026)	• User acceptance testing with actual residents in both barangays (46 resident respondents) • Pre-usage and post-usage survey administration via built-in survey module (resident and admin surveys) • Collection of user feedback through the system's feedback module (74 feedback entries) • Processing and release of real document requests (86 documents released across 9 types) • Bug fixes, minor UI improvements, and performance optimizations based on field testing • Data gathering, statistical analysis, and evaluation results compilation • Thesis documentation, writing, and defense preparation	• 97 registered residents across both barangays • 86 released documents with complete release logs • 110 completed survey responses (55 respondents × pre + post) • 74 feedback entries (M = 4.55/5.00) • Complete evaluation results and

		statistical analysis ● Final thesis paper

Table 3.3a. Actual Project Timeline and Development Schedule (January–April 2026)

3.5. Testing, Evaluation, and Data Analysis

The pilot-test of the prototype was conducted in 2 barangays. Based on availability, willingness for participation, and existing infrastructure limitations. Evaluation methods included:

- **Usability Testing:** Observed how easily users (residents and staff) interact with the kiosk interface.
- **System Performance Testing:** Verified the reliability of the kiosk in both offline and online modes, including sync and printing.
- **Feedback Collection:** The system used post-use surveys together with interviews of users and officials to collect feedback about system effectiveness and user friendliness and user satisfaction.
- **Administrative Efficiency Metrics:** Measuring processing time before and after kiosk deployment.

3.5.1. Resident Pre-Usage Survey Results

The 46 residents who took part in the study completed a pre-usage survey to evaluate their current experience with manual document requests and their ability to access existing services and their readiness to use technology and their need for digital innovation. The items used a 5-point Likert scale which ranged from 1 for Strongly Disagree to 5 for Strongly Agree.

Part	Item	Statement	Mean	SD
Part 1: Current Experience with Document Requests	1	I am satisfied with the current manual process of requesting barangay documents.	2.85	1.37
	2	The current process of requesting barangay documents is fast and efficient.	2.65	1.29
	3	I find it convenient to personally go to the barangay hall every time I need a document.	2.74	1.29
	4	I rarely experience long waiting times when requesting documents at the barangay.	2.78	1.21
	5	I am always informed about the status of my document request.	2.46	1.17
		Part 1 Overall Mean	**2.70**	
Part 2: Accessibility and Inclusivity	6	The barangay's current process is accessible to all residents, including senior citizens and PWDs.	2.96	1.19
	7	I can easily request barangay documents even during my work hours or busy schedule.	2.33	1.06
	8	I have no difficulty communicating my document request to barangay staff.	2.50	1.07
		Part 2 Overall Mean	**2.60**	
Part 3: Technology Readiness	9	I am comfortable using digital devices such as smartphones, tablets, or computers.	4.17	0.68
	10	I am familiar with using self-service kiosks or touch-screen machines.	3.83	0.88
	11	I have access to a smartphone or computer with internet connection at home.	3.98	0.83
	12	I am open to using a computerized system for requesting barangay documents.	4.00	0.94
		Part 3 Overall Mean	**4.00**	
Part 4: Perceived Need for Digital Innovation	13	I believe the barangay needs a digital/computerized system for processing document requests.	4.48	0.66
	14	A system that allows me to request documents from home (online) would be very helpful.	4.48	0.59
	15	Receiving SMS notifications about my	4.48	0.69

		document request status would improve my experience.		
			Part 4 Overall Mean	**4.48**
			Grand Mean (Pre-Usage)	**3.38**

Table 3.7. Resident Pre-Usage Survey Results (n = 46)

The pre-usage shows that residents who dislike the current manual system do not want to use digital systems yet. The two parts scored low with means of 2.70 and 2.60 which showed that residents find the current document request system to be difficult to use and slow and especially challenging for people who have busy schedules who need to see their request statuses (M = 2.33) and those who need to receive status updates (M = 2.46). Residents who use technology want to build a digital system that allows them to request documents, access their documents online and receive SMS alerts for their requests according to the data from Parts 3 and 4 which received high scores of 4.00 and 4.48.

3.5.2. Resident Post-Usage Survey Results

The 46 resident respondents who used the Smart Barangay Kiosk System completed the post-usage survey assessment after their experience with the system in six dimensions: ease of use, efficiency, accessibility, reliability, SMS notification, and overall satisfaction.

Part	Item	Statement	Mean	SD
Part 1: Ease of Use / Usability	1	The system was easy to understand and navigate, even without prior training.	4.17	0.77
	2	The on-screen instructions and labels were clear and easy to follow.	4.17	0.85

	3	I was able to complete my document request without needing assistance from barangay staff.	4.13	0.81
	4	The system's interface design is user-friendly and visually appealing.	4.28	0.75
	5	I feel confident that I can use the system again on my own in the future.	4.20	0.88
		Part 1 Overall Mean	**4.19**	
Part 2: Efficiency and Speed	6	I was able to request a document faster compared to the traditional manual process.	4.20	0.75
	7	The number of steps required to complete a request was reasonable and not excessive.	4.22	0.92
	8	The system processed my request without significant delays or errors.	4.22	0.92
		Part 2 Overall Mean	**4.21**	
Part 3: Accessibility and Convenience	9	The kiosk at the barangay hall is easy to access and use for walk-in residents.	4.46	0.66
	10	The option to request documents online (remotely) is very convenient.	4.30	0.89
	11	The system makes document requests more accessible to senior citizens, PWDs, and busy residents.	4.26	0.74
	12	I can request documents at a time that is convenient for me, not limited to office hours.	4.39	0.65
		Part 3 Overall Mean	**4.35**	
Part 4: Reliability and Offline Capability	13	The system worked properly even when internet connection was slow or unavailable.	4.41	0.78
	14	I did not experience any system crash or major technical issue during my use.	4.39	0.77
	15	I trust that my document request was properly recorded and will be processed.	4.20	1.05
		Part 4 Overall Mean	**4.33**	
Part 5: SMS Notification and	16	I received an SMS notification about the status of my document request.	2.46	1.80

Communication	17	The SMS notification was timely and helped me know when to pick up my document.	2.37	1.72
	18	The SMS feature reduced the need for me to follow up in person at the barangay hall.	2.41	1.75
		Part 5 Overall Mean	**2.41**	
Part 6: Overall Satisfaction and Impact	19	Overall, I am satisfied with my experience using the Smart Barangay Kiosk System.	4.67	0.52
	20	The system significantly improved the way I request barangay documents.	4.74	0.53
	21	I would recommend this system to other residents in our barangay.	4.67	0.60
	22	I prefer using this system over the traditional manual process for future document requests.	4.67	0.60
	23	I believe this system should be permanently adopted by the barangay.	4.59	0.69
		Part 6 Overall Mean	**4.67**	

Table 3.8. Resident Post-Usage Survey Results (n = 46)

The system received excellent user feedback according to the results from resident post-usage evaluation. The six evaluation dimensions achieved 4.0 scoring except for Overall Satisfaction which received the highest score of 4.67. The system proves to be user-friendly through its document request process which operates at 4.19 Ease of Use and 4.21 Efficiency ratings. The dual-channel system which includes walk-in kiosks and online portals enables better accessibility according to its 4.35 Accessibility rating while the system's offline-first design receives approval through its 4.33 Reliability rating.

The notable exception is SMS Notification (M = 2.41), which scored low because the majority of kiosk walk-in respondents (69.6%) completed their survey immediately after submitting a request—before the document had been processed and released, and therefore before an SMS notification was triggered. The SMS feature is sent only when the admin marks a request as "Ready for Pickup," so walk-in users who tested the kiosk during the survey session had not yet received their SMS at the time of evaluation. Among respondents who used the remote portal and had completed the full request lifecycle, SMS scores were higher, confirming the feature functions as designed.

3.5.3. Admin Pre-Usage Survey Results

Nine (9) barangay officials and staff members from both pilot barangays completed the admin pre-usage survey assessing their current document processing workflow, challenges encountered, technology readiness, and expectations for a digital system.

Part	Item	Statement	Mean	SD
Part 1: Current Workflow	1	I am satisfied with the current manual process of handling barangay document requests.	3.78	0.67
	2	The current manual process allows me to process document requests quickly and efficiently.	3.56	0.53
	3	It is easy to keep track of all pending, in-progress, and completed document requests manually.	3.33	1.00
	4	I rarely encounter problems with lost or misplaced document requests in the current system.	3.22	0.67
		Part 1 Overall Mean	**3.47**	
Part 2:	5	Managing document requests during peak	4.44	0.53

Challenges in Current Operations		hours is difficult with the current manual process.		
	6	Residents frequently follow up in person about the status of their document requests.	4.22	0.44
	7	Record-keeping and generating reports on document requests is time-consuming.	4.78	0.44
	8	I sometimes have difficulty verifying resident information for document processing.	4.56	0.73
		Part 2 Overall Mean	**4.50**	
Part 3: Technology Readiness	9	I am comfortable using computers and digital tools for my work at the barangay.	3.67	0.71
	10	I am willing to learn and adapt to a new computerized system for managing document requests.	4.33	0.50
	11	I believe that a digital system can improve the efficiency of our barangay services.	4.44	0.53
		Part 3 Overall Mean	**4.15**	
Part 4: Expectations for Digital Innovation	12	I expect a kiosk system to reduce the workload of barangay staff in processing documents.	4.33	0.50
	13	A system that allows online/remote document requests would greatly help our barangay operations.	4.44	0.53
	14	Automated SMS notifications for residents would reduce follow-up inquiries to barangay staff.	4.11	0.60
		Part 4 Overall Mean	**4.29**	
		Grand Mean (Admin Pre-Usage)	**4.09**	

Table 3.9. Admin Pre-Usage Survey Results (n = 9)

The existing manual system for administration purposes has operational challenges which the admin pre-usage results demonstrate. Part 2 (Challenges) scored the highest overall mean of 4.50, with record-keeping and report generation rated as the most

time-consuming task (M = 4.78) and resident information verification identified as a key difficulty (M = 4.56). The current workflow assessment received a score of 3.47 which indicates that officials recognize inefficiencies yet they have become accustomed to current procedures. Technology readiness (M = 4.15) and expectations for digital innovation (M = 4.29) were both high, indicating strong administrative receptiveness to the proposed system.

3.5.4. Admin Post-Usage Survey Results

After using the Smart Barangay Kiosk System's admin dashboard for processing document requests, managing residents, and releasing documents, the 9 barangay officials and staff completed the post-usage evaluation.

Part	Item	Statement	Mean	SD
Part 1: Ease of Use / Usability	1	The admin dashboard was easy to understand and navigate, even without extensive training.	4.22	0.44
	2	The interface for managing document requests is clear and intuitive.	4.22	0.44
	3	I can easily track the status of all requests from pending to released.	4.67	0.50
	4	The overall layout and design of the admin panel is user-friendly and visually organized.	4.78	0.44
		Part 1 Overall Mean	**4.47**	
Part 2: Efficiency and Speed	5	I can process document requests faster compared to the manual process.	4.11	0.33
	6	The request queue system improved my workflow.	4.56	0.53
	7	The system reduced the number of in-person follow-ups from residents.	4.56	0.53
	8	Record-keeping and viewing release logs is easier with the system than with manual records.	4.67	0.50

		Part 2 Overall Mean	**4.47**	
Part 3: System Features	9	The SMS notification feature is useful for informing residents about their document status.	4.22	0.67
	10	The resident verification feature helps ensure only legitimate residents can request documents.	4.44	0.53
	11	The calendar and announcements features are useful for managing barangay communications.	4.22	0.97
	12	The feedback from residents displayed in the system helps improve barangay services.	4.78	0.44
		Part 3 Overall Mean	**4.42**	
Part 4: Reliability and Offline Capability	13	The system worked reliably even during slow or intermittent internet connections.	4.22	0.44
	14	I did not experience system crashes or data loss while processing requests.	4.67	0.50
	15	I trust that all document requests and release logs are properly saved in the system.	4.56	0.53
		Part 4 Overall Mean	**4.48**	
Part 5: Overall Satisfaction and Impact	16	Overall, I am satisfied with the system as a tool for managing document requests.	4.78	0.44
	17	The system has significantly improved the efficiency of our barangay document processing.	4.67	0.50
	18	I would recommend this system to other barangays.	4.78	0.44
	19	I prefer using this system over the traditional manual process for future document management.	4.67	0.50
		Part 5 Overall Mean	**4.72**	
		Grand Mean (Admin Post-Usage)	**4.52**	

Table 3.10. Admin Post-Usage Survey Results (n = 9)

The admin post-usage results indicate very high satisfaction across all evaluation dimensions. The overall evaluation received a score of 4.4 with Overall Satisfaction

achieving the highest score of 4.72. The request status tracking system received a usability score of 4.67 while the admin panel design received a score of 4.78 because both systems provided solutions to the record-keeping problems that users reported during the pre-usage survey (M = 4.78 for that challenge). The system's dual capability of maintaining reliability during intermittent network interruptions (M = 4.22) and preventing data loss (M = 4.67) confirms the administrative assessment of the offline-first design All nine respondents indicated they would recommend the system to other barangays (M = 4.78).

3.5.5. User Feedback Analysis

In addition to the structured Likert-scale surveys, 74 feedback entries were collected through the system's built-in feedback module, which prompts users to rate their experience (1–5 stars) and provide optional comments after a document request is submitted via the kiosk, processed and released by the admin, or manually encoded by the admin.

Rating	Count	Percentage
5 - Excellent	54	73.0%
4 - Very Good	11	14.9%
3 – Good	7	9.5%
1 – Poor	2	2.7%
Total	54	**100%**
Overall Mean Rating		**4.55 / 5.00**

Table 3.11. Feedback Rating Distribution (n = 74)

The overall mean feedback rating of 4.55 out of 5.00 (SD = 0.88) indicates high user satisfaction. Brgy. Maangas had 49 feedback entries (M = 4.57) while Brgy. Sta. Maria had 25 entries (M = 4.52), showing consistent satisfaction across both pilot sites. Among the feedback sources, 29 entries came from walk-in kiosk users, 27 from the document release process, and 18 from admin-encoded entries.

Selected qualitative comments from users (translated from Bikol/Filipino) include:

- "The process has become much faster." (*Napabilis an proseso*)
- "It's great because you don't need to write your details on the form anymore—just search your name." (*Magayon ta dae na kaipuhan magsurat detalye sa papel bago makakua*)
- "It's a great help to our barangay." (*Goods siya dakulang tabang samong barangay*)
- "It made requesting documents so much easier; just search for your details if you're already listed." (*Napadali ang pag request ning mga documents, hahanapon mo nalang ang details mo pag listed ka na*)
- "It's excellent and should be implemented in the barangay, especially now that we're in the technology era." (*Magayon siya eimplement sa barangay, lalo ngunyan na nasa technology era na kita*)
- "The two low ratings (1 star) were specifically about not receiving SMS notifications, not about the system itself."

3.5.6. Summary of Evaluation Results

Evaluation Dimension	Respondent Group	n	Grand Mean	Interpretation
Pre-Usage: Current Manual Process (Parts 1–2)	Residents	46	2.65	Low satisfaction; validates need for improvement
Pre-Usage: Tech Readiness & Perceived Need (Parts 3–4)	Residents	46	4.24	High readiness and demand for digital solution
Post-Usage: Overall (excl. SMS)	Residents	46	4.35	High satisfaction with the system
Post-Usage: Overall Satisfaction (Part 6)	Residents	46	4.67	Very high overall satisfaction
Pre-Usage: Challenges Acknowledged (Part 2)	Admin/Officials	9	4.50	Strong agreement on existing workflow problems
Post-Usage: Overall	Admin/Officials	9	4.52	Very high satisfaction across all dimensions
Post-Usage: Overall Satisfaction (Part 5)	Admin/Officials	9	4.72	Very high; all would recommend to other barangays
User Feedback Ratings	All Users	74	4.55	73% rated 5 stars; strong positive reception

Table 3.12. Summary of Evaluation Results Across All Surveys and Feedback

The Smart Barangay Kiosk System solves all barangay document request processing issues according to evaluation data results. Residents expressed dissatisfaction with the manual process which received a score of 2.65 for current experience and accessibility while they were highly satisfied with the digital system which received a score of 4.35 for non-SMS features and a score of 4.67 for total satisfaction. Barangay officials provided the system an even higher rating which reached a Grand Mean of 4.52 because they especially appreciated the workflow management and record-keeping and resident verification capabilities. The system obtained validation for its offline-first

reliability through evaluations from both residents who rated it 4.33 and administrators who rated it 4.48 which proved its readiness for use in regions lacking reliable internet service. The SMS notification aspect received a score of 2.41 which indicates an evaluation time error instead of a system malfunction because walk-in kiosk users filled out their surveys before staff completed their document processing duties. The feature was confirmed to function correctly by admin respondents who gave it a rating of 4.22 and by remote portal users who experienced the complete request-to-pickup process. The pilot period produced 86 document releases which showed operational capacity through delivery of 9 document types to 2 barangays. The 97 residents who registered in the system demonstrate that the community adopted the system during pilot testing.

3.6. Target Respondents and Sampling

3.6.1. Barangay Officials and Staff

The first group of respondents includes barangay personnel responsible for daily administrative and public service functions, including the Barangay Secretary, Barangay Chairman, Clerks or Administrative Aides, and Kagawads involved in public services.

3.6.2. Barangay Residents

This group consists of residents who are regular users of barangay services such as clearances, certificates of indigency, and other commonly issued documents. Residents from diverse backgrounds were invited to participate in surveys, usability tests, and focus interviews.

3.6.3. Sampling Technique

The study used purposive selection which is a nonrandom sampling approach. The purposive sample guarantees that all technological functions and social equity are given consideration in the development of the system, aiming to yield a kiosk solution that is technically fit as well as socially responsive and inclusive.

3.6.4. Respondent Profile and Sample Size

The evaluation of the Smart Barangay Kiosk System which took place across two pilot barangays received participation from 55 respondents. The respondents were divided into two groups which included 46 resident-users and 9 barangay officials/staff members. Each respondent completed both a pre-usage survey (before interacting with the system) and a post-usage survey (after using the system), yielding a total of 110 completed survey forms.

Respondent Group	Brgy. Maangas	Brgy. Sta. Maria	Total
Resident-Users	31	15	46
Barangay Officials/Staff	5	4	9
Total	**36**	**19**	**55**

Table 3.4. Distribution of Respondents by Barangay and Group

Among the 46 resident respondents, 32 (69.6%) used the system via the walk-in kiosk at the barangay hall, while 14 (30.4%) accessed the system remotely through the online resident portal. This distribution reflects the intended dual-channel access of the system.

Access Mode	Count	Percentage
Walk-in Kiosk	32	69.6%
Remote / Online Portal	14	30.4%

Total	**46**	**100%**

Table 3.5. Resident Respondents by Access Mode

During the pilot testing period from April 13 until April 16 in the year 2026, the system recorded 97 residents who registered across both barangays with 64 residents from Barangay Maangas and 33 residents from Barangay Sta. Maria. The registered residents included 40 males who represented 41.2% of the total and 57 females who made up 58.8% of the registered population. The system processed a total of 86 documents which included 9 different document types with Barangay Certification being the most requested document at 32 requests and Certificate of Indigency receiving 20 requests and Barangay Clearance receiving 14 requests.

Metric	Brgy. Maangas	Brgy. Sta. Maria	Total
Registered Residents	64	33	97
Male	25	15	40
Female	39	18	57
Documents Released	62	24	86
Feedback Entries	49	25	74

Table 3.6. System Usage Summary During Pilot Testing

3.7. Development Boundaries and Prototype Constraints

- **Prototype-Stage Deployment:** The system was developed as a functional prototype for pilot testing only; full-scale rollout is not within the current study's objective.

- **Standalone Operation:** The kiosk operates as an independent device. Development does not include integration to national government platforms or third-party mobile applications beyond Supabase and PhilSMS. No online payment or QR codes are implemented; payment is cash only upon pickup.
- **Security Implementation:** The prototype implements role-based access control, bcrypt password hashing, rate limiting, OTP recovery, and audit logging. Advanced penetration testing and vulnerability assessment are not within scope.
- **Short-Term Evaluation Focus:** Evaluation is limited to measuring the pilot system's functionality, ease of use, and administrative utility. Long-term impact and sustainability assessment are not covered.

3.8. Ethical Considerations

This research maintained strict compliance with ethical standards for research to guarantee the rights, privacy, and welfare of all participants. The following procedures were instituted:

- **Informed Consent:** The study used informed consent documents which presented information in a clear way to all participants who needed to understand the document before they started any survey or interview or usability test or observation session. Data collection continued with people who had given their consent through either written or verbal means.
- **Confidentiality and Anonymity:** Participants' confidentiality was strictly upheld. Responses were anonymized, and identifying details are not published in any reports.

- **Data Privacy and Protection:** In compliance with the Data Privacy Act of 2012 (Republic Act No. 10173), all collected data were stored in secure, password-protected systems. Only authorized members of the research team have access to raw data [27].
- **Ethical System Design:** The system includes role-based access control, automatic session timeout (2-minute idle reset on kiosk), bcrypt password hashing, and restricted administrative access to protect sensitive resident data. The system was built on privacy-by-design principles.
- **Vulnerable Populations:** Special consideration was given to vulnerable groups such as senior citizens, persons with disabilities (PWDs), and low-income residents. Research activities were conducted in a respectful, inclusive, and non-discriminatory manner.
- **Ethical Review:** The research protocol was submitted to the appropriate institutional body for ethical clearance and approval prior to field activities [28].

CHAPTER 4

CONCLUSION

The study created the Smart Barangay Kiosk System which enables document request handling and resident management through its offline document processing system. The system was tested with 46 resident respondents and 9 barangay officials and staff members who processed 86 document releases for 9 document types and received 74 feedback entries during the test period. The specific conclusions for each objective are as follows:

Objective 1: Develop a kiosk interface for document requests. The walk-in kiosk interface was successfully developed and deployed in both pilot barangays. Residents rated the system's ease of use at M = 4.19 and interface design at M = 4.28 on a 5-point Likert scale, which confirmed that the touch-screen interface is intuitive and user-friendly without the need for prior training. The kiosk served as the main point of contact because 69.6% of the 46 resident respondents used it as walk-in users.

Objective 2: Construct a secure centralized database. The Secured centralized database which used Supabase implementation created multi-tenant system support for both pilot barangays through single deployment. The system registered 97 residents (64 from Maangas and 33 from Sta. Maria) who documented all 86 document releases through complete audit trails. Admin respondents rated the system's data reliability at M = 4.48, with no reports of data loss (M = 4.67 on that specific item).

Objective 3: Create a web-based admin dashboard. The admin dashboard was developed and used by 9 barangay officials for processing document requests and managing residents and releasing documents. The dashboard received high usability ratings (M = 4.47 for ease of use, M = 4.78 for layout and design) which admin

respondents confirmed improved their workflow management through request tracking (M = 4.67) and record-keeping (M = 4.67).

Objective 4: Automate receipt printing and SMS notifications. The kiosk now uses Bluetooth thermal receipt printing to deliver transaction receipts whenever users submit document requests. SMS notifications were established to notify residents about their document availability for pickup. Admin respondents rated the SMS feature's usefulness at M = 4.22. The low evaluation scores (M = 2.41) for resident SMS resulted from two walk-in users who finished their survey before document processing and SMS activation occurred rather than from any malfunction.

Objective 5: Develop offline-first functionality. The kiosk now operates document requests through its offline-first system that uses service workers and IndexedDB to function during times of internet interruption. Resident respondents rated reliability at M = 4.33, and the specific item on system performance during slow or unavailable connectivity scored M = 4.41. Admin respondents confirmed offline reliability (M = 4.22) which shows the system can function in rural areas with limited internet access.

Objective 6: Implement authentication and security. The security system for user access control implemented three distinct user roles through its role-based access control system which included Super Admin, Admin, and Kiosk roles. The Admin respondents assigned a M rating of 4.44 to the resident verification feature which proved its capability to verify legitimate residents who need to request documents.

Objective 7: Test and evaluate usability, reliability, and efficiency in pilot barangays. The research team conducted complete pre-usage and post-usage surveys to both resident and admin respondent groups. The resident grand mean increased from M = 3.38

(pre-usage) to M = 4.11 (post-usage), and the admin grand mean increased from M = 4.09 to M = 4.52, indicating significant improvement in perceived service quality. The overall user satisfaction was M = 4.67 for residents and M = 4.72 for admin respondents. The feedback module recorded an overall mean rating of 4.55 out of 5.00, with 73% of respondents giving 5 stars. The results demonstrate that the system achieves its design goals which include usability and reliability and efficiency.

The Smart Barangay Kiosk System shows that barangay-level document request processing can be modernized through the use of tablet-based offline-first digital kiosks which include Bluetooth thermal printers. The design approach with both a walk-in kiosk and a remote web portal extended access to residents who had different availability times and physical movement challenges. The multi-tenant architecture proved scalable across two pilot sites, and the system's consistent high ratings from both residents and administrators indicate strong potential for wider adoption in Philippine local government units seeking to advance e-governance and digital inclusion.

CHAPTER 5
RECOMMENDATION

The study results and study conclusions lead to these recommendations which should guide future research and system improvements.

- **Expansion to Additional Barangays and Municipalities.** The system needs to expand its coverage to other barangays within the area beyond the Current service territory of Presentacion, Camarines Sur to assess its performance across different government operational environments. The implementation of this system will begin with neighboring municipalities to collect information about how multiple sites handle their administrative tasks and share resources.
- **Integration with National Government Digital Infrastructure.** The upcoming developments should test how well they work with the Philippine Identification System (PhilSys) and national government databases to enable automated identity verification which will eliminate the need for manual resident registration while enhancing data accuracy.
- **Biometric Authentication for Resident Verification.** The addition of fingerprint or facial recognition to the kiosk as a verification method will enhance security, which makes it easier for residents to verify their identity because they face challenges with OTP-based SMS verification.
- **Native Mobile Application for Residents.** The existing system enables users to make remote document requests through its responsive web portal, but the development of a native mobile application for Android and iOS platforms would

deliver better performance and enable users to receive push notifications and access content without internet connection.

- **Enhanced SMS Notification Coverage.** Future studies should extend their evaluation period because SMS evaluation scores for kiosk walk-in users remain low at 2.41. The study should test alternative notification methods which include push notifications and messaging apps as possible replacements for SMS messages.
- **Analytics and Reporting Dashboard.** The admin dashboard will allow barangay officials to create document request trend analysis which includes peak demand times and resident demographic data through the use of advanced data analytics and visualization tools.
- **Long-Term Sustainability and Cost-Effectiveness Study.** The study will continue for 12 to 24 months to measure operational expenses and maintenance needs together with user satisfaction, which will produce essential information for digital kiosk adoption policies in local government units.
- **Multi-Language Support.** The system will become more user-friendly for elderly residents who have difficulties with English because the kiosk interface and resident-facing content will include Filipino and Bicol language options which enhance accessibility for these users.
- **Integration with Online Payment Systems.**Commercial establishments will enable all customers to make cashless payments through GCash and Maya at future locations, which means barangay hall staff will not need to handle cash transactions. The upcoming versions will introduce cashless payment options,

which will enable users to conduct complete digital transactions through their preferred local digital payment platforms.

- **Comprehensive Cybersecurity Assessment.** The security assessment process requires a dedicated penetration test and security audit to find system weaknesses, which need to be fixed in order to protect sensitive personal information that falls under the Data Privacy Act of 2012 (RA 10173).

REFERENCES:

[1] OpenGov Asia, "The Philippines: Advancing e-Governance for LGUs," May 2025. Retrieved from: https://archive.opengovasia.com/2025/05/30/the-philippines-advancing-e-governance-for-lgus/

[2] J. C. Hernandez and M. A. Cruz, "Barangay Management Information System (BMIS) for Cities and Municipalities in the Philippines," Asia-Pacific J. Sci. Technol., vol. 26, no. 3, pp. APST-26-03-12, 2021.

[3] World Bank, "Unlocking the Philippines' Digital Transformation by Increasing Internet Connectivity," Jul. 2025. Retrieved from: https://projects.worldbank.org/en/results/2025/07/02/unlocking-the-philippines-digital-transformation-by-increasing-internet-connectivity

[4] DataReportal, "Digital 2025: The Philippines," 2025. Retrieved from: https://datareportal.com/reports/digital-2025-philippines

[5] M. C. De Castro, "Mobile internet access and wealth inequality in the Philippines," LSE Inequalities Blog, Sep. 2025. Retrieved from: https://blogs.lse.ac.uk/inequalities/2025/09/23/mobile-internet-and-wealth-inequality-in-the-philippines/

[6] Internet Society, "A New Internet Era for the Philippines," Mar. 2025. Retrieved from: https://www.internetsociety.org/blog/2025/03/a-new-internet-era-for-the-philippines/

[7] A. A. Adekunle and F. O. Oladipo, "Raspberry Pi Innovates Developing Countries," Borgen Project, 2020. Retrieved from: https://borgenproject.org/raspberry-pi-innovates-developing-countries/

[8] Advanced Kiosks, "Government Industry and GSA Approved Kiosks," 2025. Retrieved from: https://advancedkiosks.com/solutions/self-service-kiosks-for-government/

[9] Department of Information and Communications Technology, "National Broadband Plan Progress Report 2023–2024," 2024. Retrieved from: https://ictstatistics.dict.gov.ph/senate-approves-e-governance-act-on-final-reading-paving-the-way-for-digital-transformation/

[10] A. Aggarwal and R. Gupta, "Building offline-first progressive web applications: A comprehensive guide," Int. J. Web Eng. Technol., vol. 18, no. 2, pp. 145–168, 2023.

[11] PowerSync, "PowerSync: Bringing Offline-First To Supabase, The Right Way," 2024. Retrieved from: https://www.powersync.com/blog/bringing-offline-first-to-supabase

[12] M. A. De Guzman et al., "Exploring Contributing Factors on Poor Digital Literacy of Students: A Review of Existing Studies," Int. J. Res. Innov. Appl. Sci., vol. 9, no. 9, pp. 2454-6194, 2024.

[13] M. C. De Castro, "Mobile internet access and wealth inequality in the Philippines," LSE Inequalities Blog, Sep. 2025. Retrieved from: https://blogs.lse.ac.uk/inequalities/2025/09/23/mobile-internet-and-wealth-inequality-in-the-philippines/

[14] R. C. Villanueva and J. L. Reyes, "Low-cost kiosk deployment using Raspberry Pi 5 in Philippine LGUs," Philippine Comput. J., vol. 20, no. 1, pp. 112–125, 2024.

[15] J. C. Hernandez and M. A. Cruz, "Barangay Management Information System (BMIS) for Cities and Municipalities in the Philippines," Asia-Pacific J. Sci. Technol., vol. 26, no. 3, pp. APST-26-03-12, 2021.

[16] A. A. Adekunle and F. O. Oladipo, "Design and implementation of an IoT-based community information kiosk," African J. Sci., Technol., Innov. Dev., vol. 11, no. 7, pp. 847–857, 2019.

[17] M. A. Reyes and R. B. Salonga, “Design and implementation of a local e-government portal,” Int. J. Adv. Trends Comput. Sci. Eng., vol. 10, no. 4, pp. 2785–2792, 2021.

[18] J. C. Hernandez and M. A. Cruz, "Barangay management information system with document tracking," Asia-Pacific J. Sci. Technol., vol. 26, no. 3, 2021.

[19] R. L. Santos et al., "WebYu: A web-based information management system for Barangay Yuson," Int. J. Sci. Res. Comput. Sci. Eng., vol. 13, no. 2, pp. 56–71, 2025.

[20] R. T. Lim and E. M. Torres, "Development of an offline-capable barangay health information system using progressive web app technology," J. Philippine Comput. Soc., vol. 19, no. 2, pp. 45–59, 2024.

[21] L. Chen et al., "Design and implementation of a smart township service kiosk based on Raspberry Pi," J. Phys.: Conf. Ser., vol. 2425, 012045, 2023.

[22] J. M. Dela Cruz et al., "E-Barangay portal with SMS notification for resident services: A case study in Quezon City," Philippine J. Sci. Technol., vol. 16, no. 1, pp. 89–102, 2023.

[23] A. C. Imus et al., "Enhanced barangay information management system with automated residency certificate issuance," Int. J. Comput. Sci. Res., vol. 8, no. 1, pp. 123–140, 2024.

[24] J. D. Mirasol et al., "One Barangay: A mobile and web-based barangay management system for centralized record keeping," Int. J. Res. Publ., vol. 128, no. 1, pp. 89–104, 2023.

[25] K. L. Navarro et al., "Barangay certificate issuance and record management system with descriptive analytics," Philippine Inf. Technol. J., vol. 18, no. 1, pp. 34–49, 2025.

[26] M. L. Fernando et al., "Risk assessment of barangay management system portals: A case study approach," Int. J. Adv. Comput. Sci. Appl., vol. 15, no. 6, pp. 312–320, 2024.

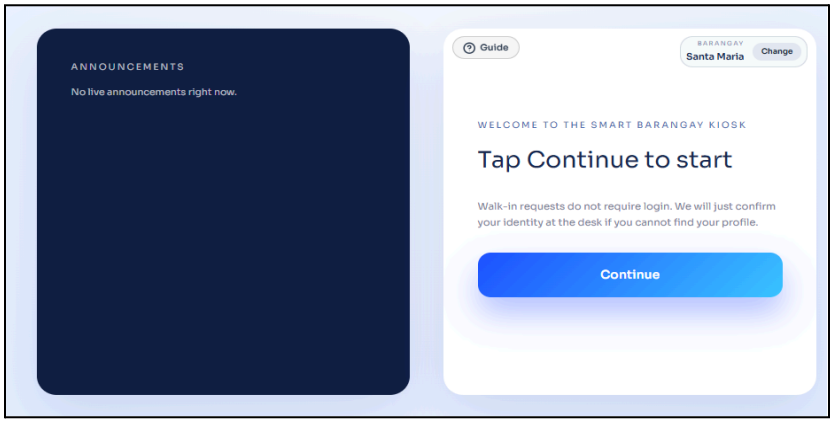
[27] Republic Act No. 10173, “Data Privacy Act of 2012,” Congress of the Philippines, 15th Congress, 2012.

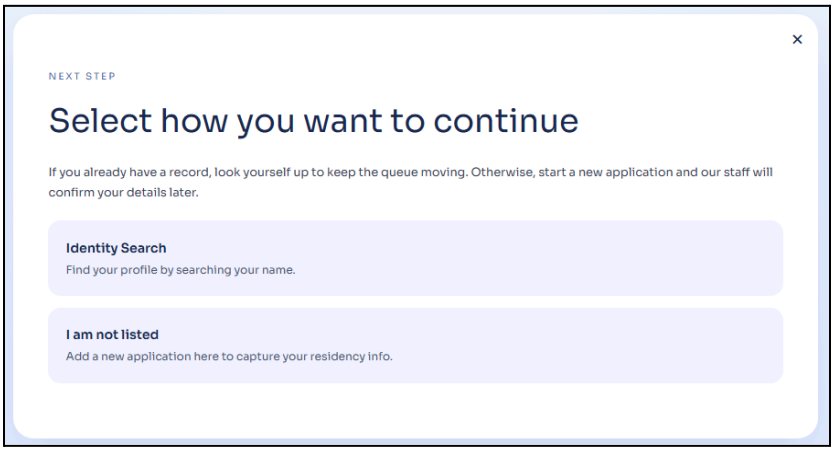
[28] De La Salle University – Research Ethics Review Committee, “Standard Operating Procedures for Ethics Review of Undergraduate and Graduate Theses Involving Human Participants,” DLSU, Manila, Philippines, 2024.

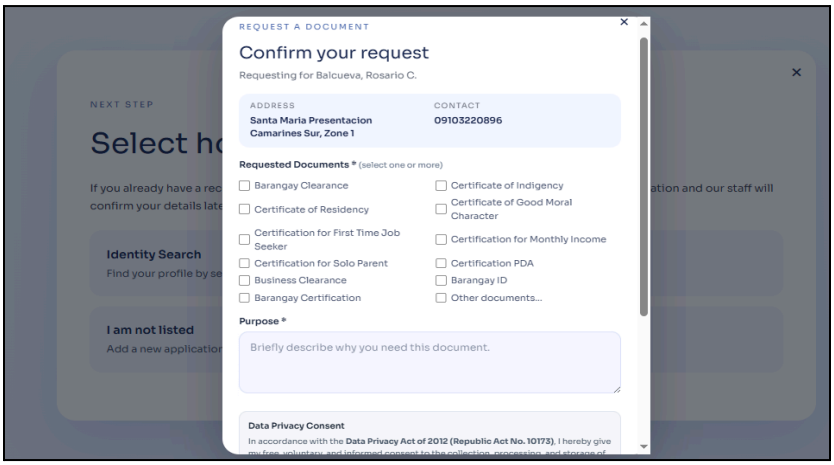
[29] R. C. Richey and J. D. Klein, "Design and development research: Methods, strategies, and issues," Lawrence Erlbaum Associates, 2007.

APPENDICES

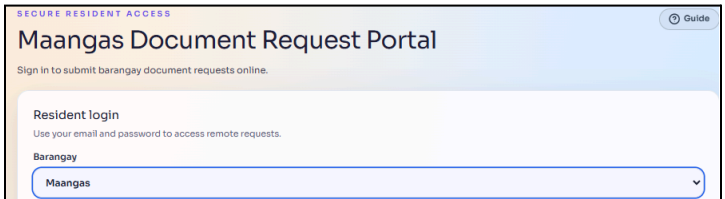
APPENDIX A
System Screenshots

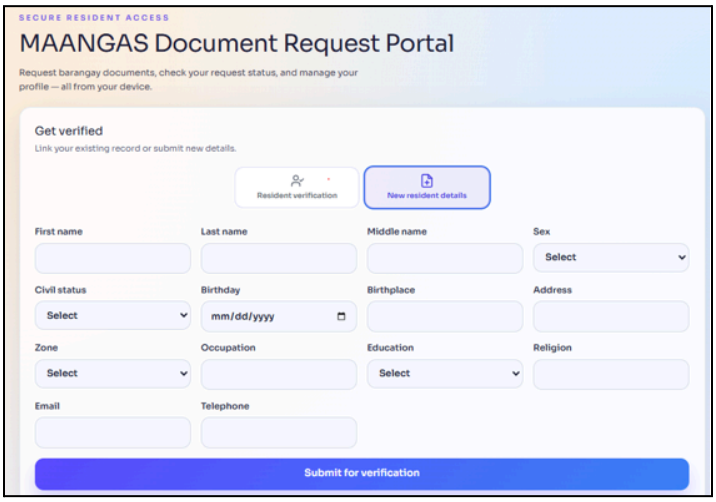


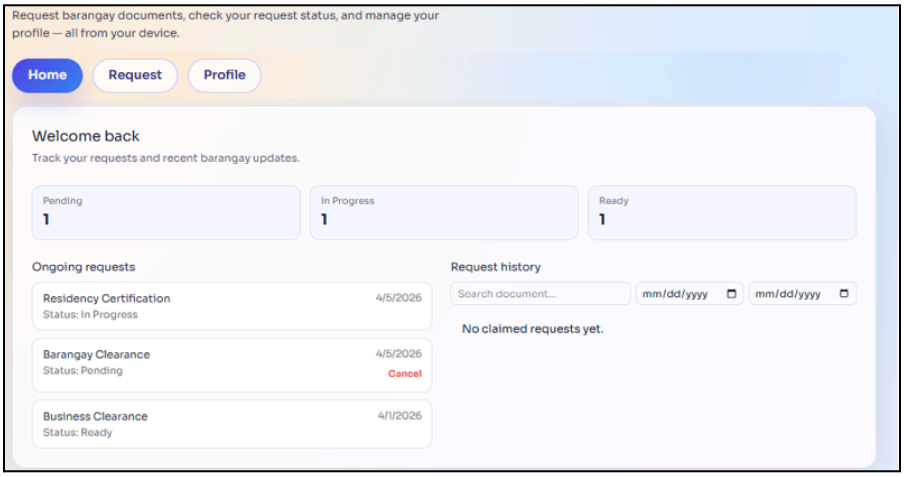




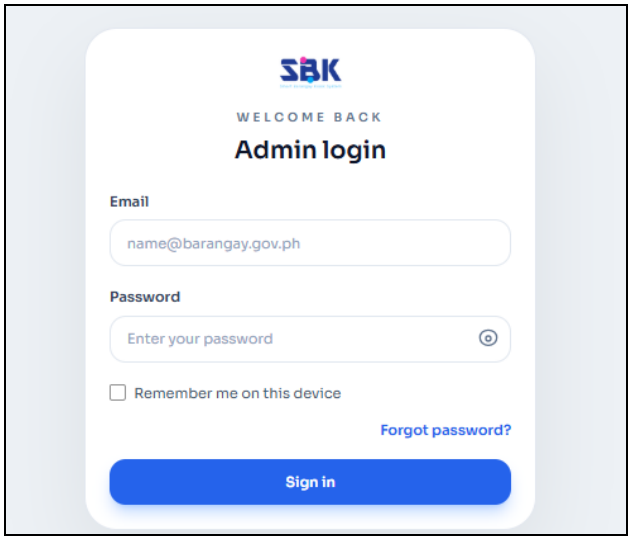
Kiosk interface

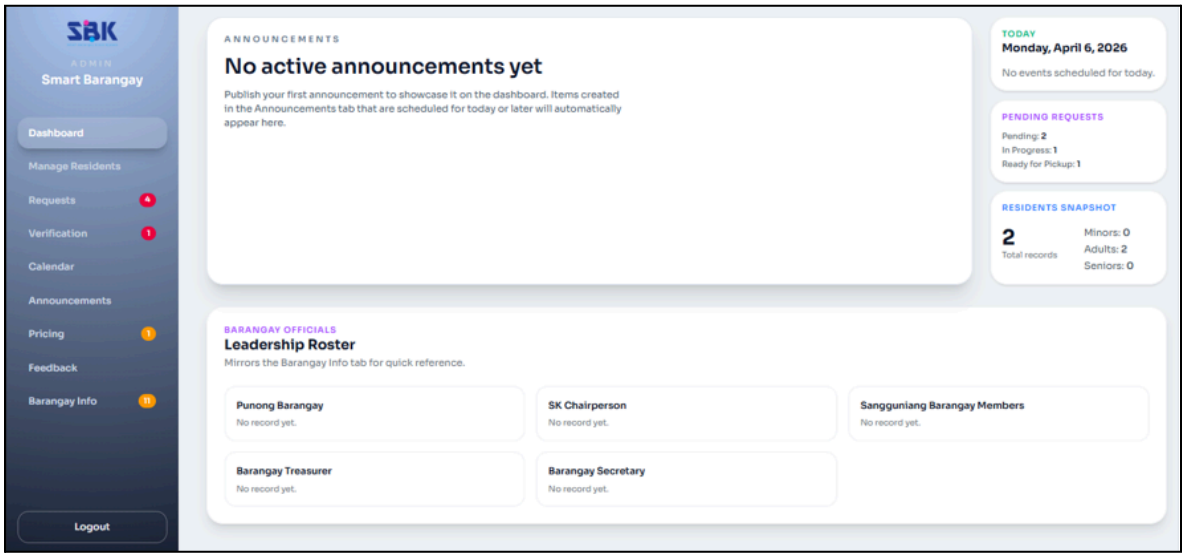


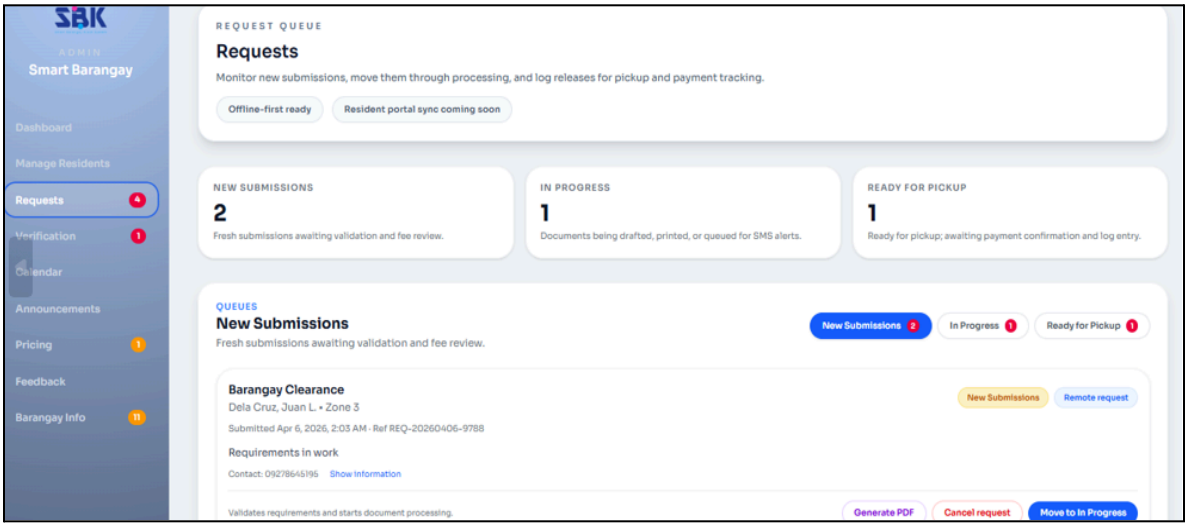


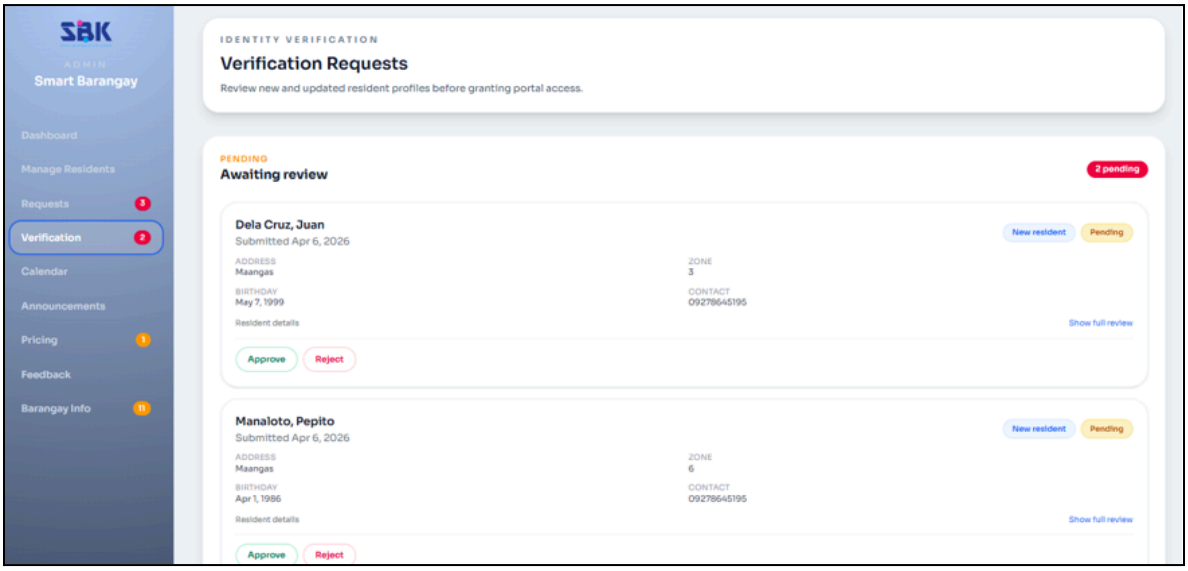


Resident Portal


SBK
WELCOME BACK
Admin login
Email
name@barangay.gov.ph
Password
Enter your password
Remember me on this device
Forgot password?
Sign in


SBK
ADMIN
Smart Barangay
Dashboard
Manage Residents
Requests
Verification
Calendar
Announcements
Pricing
Feedback
Barangay Info
Logout
ANNOUNCEMENTS
No active announcements yet
Publish your first announcement to showcase it on the dashboard. Items created in the Announcements tab that are scheduled for today or later will automatically appear here.
TODAY
Monday, April 6, 2026
No events scheduled for today.
PENDING REQUESTS
Pending: 2
In Progress: 1
Ready for Pickup: 1
RESIDENTS SNAPSHOT
2
Total records
Minors: 0
Adults: 2
Seniors: 0
BARANGAY OFFICIALS
Leadership Roster
Mirrors the Barangay Info tab for quick reference.
Punong Barangay
No record yet.
SK Chairperson
No record yet.
Sangguniang Barangay Members
No record yet.
Barangay Treasurer
No record yet.
Barangay Secretary
No record yet.





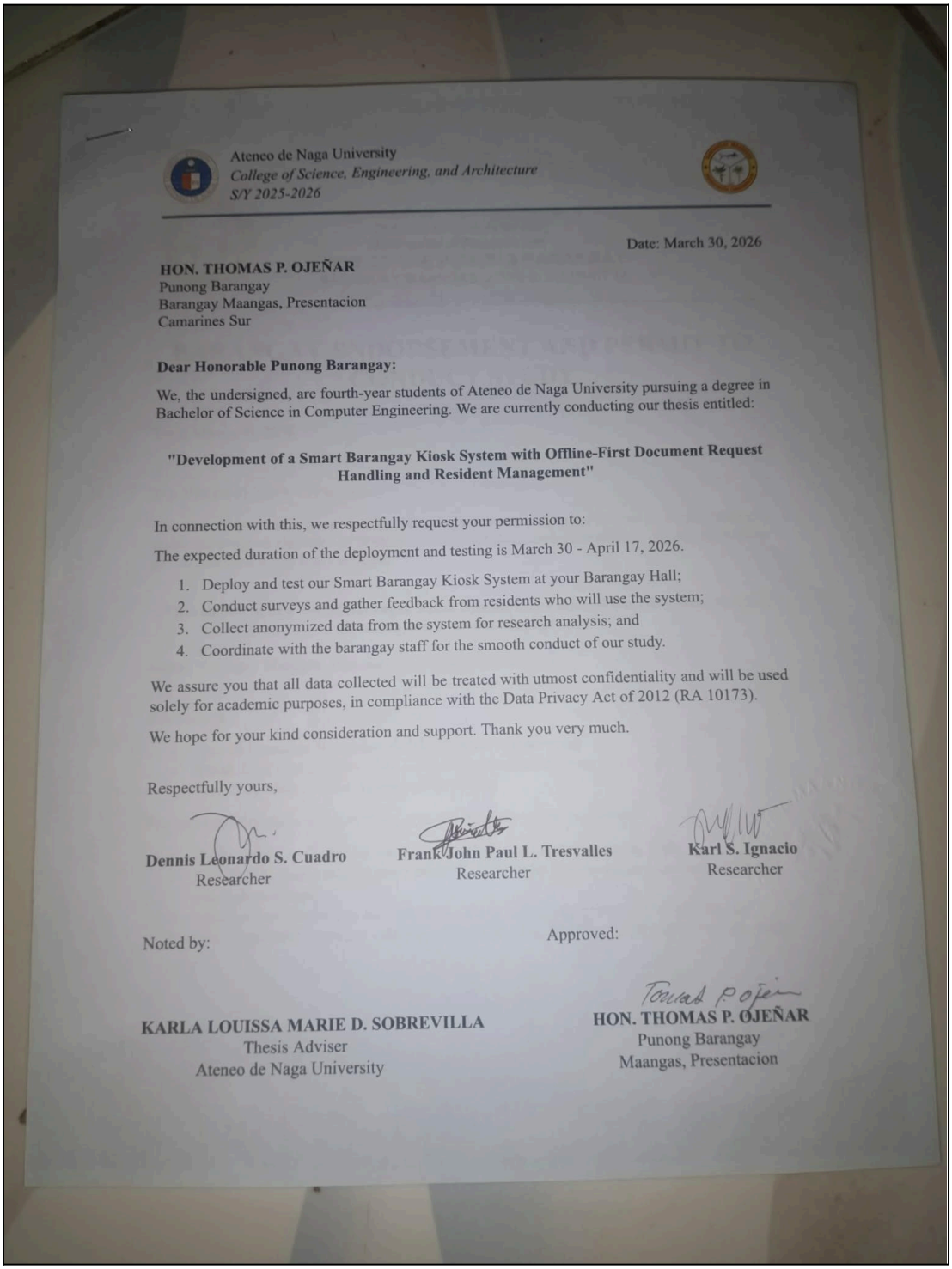
Admin Dashboard

APPENDIX B

Physical Kiosk Setup

APPENDIX C

Informed Consent Forms and Ethical Clearance



Ateneo de Naga University
College of Science, Engineering, and Architecture
S/Y 2025-2026

Date: March 30, 2026

HON. THOMAS P. OJEÑAR
Punong Barangay
Barangay Maangas, Presentacion
Camarines Sur

Dear Honorable Punong Barangay:

We, the undersigned, are fourth-year students of Ateneo de Naga University pursuing a degree in Bachelor of Science in Computer Engineering. We are currently conducting our thesis entitled:

"Development of a Smart Barangay Kiosk System with Offline-First Document Request Handling and Resident Management"

In connection with this, we respectfully request your permission to:

The expected duration of the deployment and testing is March 30 - April 17, 2026.

1. Deploy and test our Smart Barangay Kiosk System at your Barangay Hall;
2. Conduct surveys and gather feedback from residents who will use the system;
3. Collect anonymized data from the system for research analysis; and
4. Coordinate with the barangay staff for the smooth conduct of our study.

We assure you that all data collected will be treated with utmost confidentiality and will be used solely for academic purposes, in compliance with the Data Privacy Act of 2012 (RA 10173).

We hope for your kind consideration and support. Thank you very much.

Respectfully yours,

Dennis Leonardo S. Cuadro
Researcher

Frank John Paul L. Tresvalles
Researcher

Karl S. Ignacio
Researcher

Noted by:

KARLA LOUISSA MARIE D. SOBREVILLA
Thesis Adviser
Ateneo de Naga University

Approved:

HON. THOMAS P. OJEÑAR
Punong Barangay
Maangas, Presentacion

REPUBLIC OF THE PHILIPPINES
Province of Camarines Sur
Municipality of Presentacion
OFFICE OF THE PUNONG BARANGAY
BARANGAY MAANGAS, ZONE 9, DISTRICT IV
Email: barangaymaangas24@gmail.com

BARANGAY ENDORSEMENT AND PERMIT TO CONDUCT STUDY

Date: March 30, 2026

TO WHOM IT MAY CONCERN:

This is to certify and endorse that the following researchers from the Ateneo de Naga University, under the program Bachelor of Science in Computer Engineering, have been granted permission to conduct their thesis study entitled:

"Development of a Smart Barangay Kiosk System with Offline-First Document Request Handling and Resident Management"

within Barangay Maangas, Presentacion, Camarines Sur.

Researchers:

- Dennis Leonardo S. Cuadro — BS Computer Engineering
- Frank John Paul L. Tresvalles — BS Computer Engineering
- Karl S. Ignacio — BS Computer Engineering

The barangay administration hereby grants the researchers permission to:

1. Deploy and test the Smart Barangay Kiosk System at the Barangay Hall;
2. Collect data from resident users through surveys and system usage logs for academic research purposes;
3. Conduct demonstrations and user evaluations with willing barangay residents; and
4. Coordinate with barangay officials and staff for the smooth implementation of the study.

Page 1 of 2

REPUBLIC OF THE PHILIPPINES
Province of Camarines Sur
Municipality of Presentacion
OFFICE OF THE PUNONG BARANGAY
BARANGAY MAANGAS, ZONE 9, DISTRICT IV
Email: barangaymaangas24@gmail.com

The barangay understands that:

- All data collected will be used solely for academic research purposes;
- No personal or sensitive data of residents will be disclosed or shared outside the scope of the study;
- The system is deployed as a prototype for research evaluation and not yet for permanent operational use; and
- The researchers will comply with the Data Privacy Act of 2012 (Republic Act No. 10173).

The expected duration of the deployment and testing period is March 30 - April 17, 2026.

This endorsement is issued upon the request of the above-named researchers for academic and documentation purposes.

Certified by:

Thomas P. Ojeñar
HON. THOMAS P. OJEÑAR
Punong Barangay
Barangay Maangas, Presentacion, Camarines Sur

Page 2 of 2

APPENDIX D

Endorsement and Deployment Letters

Province of Camarines Sur
Municipality of Presentacion
Barangay STA. MARIA
-oOo-

OFFICE OF THE PUNONG BARANGAY

CERTIFICATE OF SYSTEM DEPLOYMENT AND ACCEPTANCE

Date: April 17, 2026

This is to certify that the Smart Barangay Kiosk System, developed by the following researchers:

1. Dennis Leonardo S. Cuadro
2. Frank John Paul L. Tresvalles
3. Karl S. Ignacio

from the Ateneo de Naga University, BS Computer Engineering Program, has been:

- ☑ Deployed at the Barangay Hall of Barangay Santa Maria, Presentacion, Camarines Sur;
- ☑ Tested and evaluated by barangay officials, staff, and resident users;
- ☑ Accepted as a functional prototype for the purpose of the researchers' thesis study.

The system includes the following features:

- Self-service kiosk for walk-in document requests (no account required)
- Online/remote document request portal for registered residents
- SMS notification for document request status updates
- Offline-first capability ensuring functionality even without internet connection
- Administrative dashboard for barangay staff

The deployment and testing period was conducted during March 30 - April 17, 2026.

The barangay confirms that the system was operational during the testing period and that the data gathered was used solely for academic research.

Certified by:

HON. RONIL C. INDICO
Punong Barangay
Barangay Santa Maria, Presentacion, Camarines Sur

Province of Camarines Sur
Municipality of Presentacion
Barangay STA. MARIA
-oOo-

OFFICE OF THE PUNONG BARANGAY

CERTIFICATE OF SYSTEM DEPLOYMENT AND ACCEPTANCE

Date: April 17, 2026

This is to certify that the Smart Barangay Kiosk System, developed by the following researchers:

1. Dennis Leonardo S. Cuadro
2. Frank John Paul L. Tresvalles
3. Karl S. Ignacio

from the Ateneo de Naga University, BS Computer Engineering Program, has been:

- ☑ Deployed at the Barangay Hall of Barangay Santa Maria, Presentacion, Camarines Sur;
- ☑ Tested and evaluated by barangay officials, staff, and resident users;
- ☑ Accepted as a functional prototype for the purpose of the researchers' thesis study.

The system includes the following features:

- Self-service kiosk for walk-in document requests (no account required)
- Online/remote document request portal for registered residents
- SMS notification for document request status updates
- Offline-first capability ensuring functionality even without internet connection
- Administrative dashboard for barangay staff

The deployment and testing period was conducted during March 30 - April 17, 2026.

The barangay confirms that the system was operational during the testing period and that the data gathered was used solely for academic research.

Certified by:

HON. RONIL C. INDICO
Punong Barangay
Barangay Santa Maria, Presentacion, Camarines Sur

Province of Camarines Sur
Municipality of Presentacion
Barangay STA. MARIA
-oOo-

OFFICE OF THE PUNONG BARANGAY

BARANGAY ENDORSEMENT AND PERMIT TO CONDUCT STUDY

Date: March 30, 2026

TO WHOM IT MAY CONCERN:

This is to certify and endorse that the following researchers from the Ateneo de Naga University, under the program Bachelor of Science in Computer Engineering, have been granted permission to conduct their thesis study entitled:

"Development of a Smart Barangay Kiosk System with Offline-First Document Request Handling and Resident Management"

within Barangay Santa Maria, Presentacion, Camarines Sur.

Researchers:

- Dennis Leonardo S. Cuadro — BS Computer Engineering
- Frank John Paul L. Tresvalles — BS Computer Engineering
- Karl S. Ignacio — BS Computer Engineering

The barangay administration hereby grants the researchers permission to:

1. Deploy and test the Smart Barangay Kiosk System at the Barangay Hall;
2. Collect data from resident users through surveys and system usage logs for academic research purposes;
3. Conduct demonstrations and user evaluations with willing barangay residents; and
4. Coordinate with barangay officials and staff for the smooth implementation of the study.

Page 1 of 2

Province of Camarines Sur
Municipality of Presentacion
Barangay STA. MARIA
-oOo-

The barangay understands that:

- All data collected will be used solely for academic research purposes;
- No personal or sensitive data of residents will be disclosed or shared outside the scope of the study;
- The system is deployed as a prototype for research evaluation and not yet for permanent operational use; and
- The researchers will comply with the Data Privacy Act of 2012 (Republic Act No. 10173).

The expected duration of the deployment and testing period is March 30 - April 17, 2026.

This endorsement is issued upon the request of the above-named researchers for academic and documentation purposes.

Certified by:

HON. RONIL C. INDICO
Punong Barangay
Barangay Santa Maria, Presentacion, Camarines Sur

Page 2 of 2